

EVERYTHING You Need to Know About Oscillators

A technical reference on linear oscillations:
from one mass on a spring to the wave equation

Technical reference for lecture slides

July 23, 2026

Abstract

This document is a self-contained technical reference on the classical harmonic oscillator, intended as source material for teaching slides. Every result is derived in full, with all intermediate steps displayed. The topics, in order: the simple harmonic oscillator (where the complex-exponential method is introduced and justified), the damped oscillator (with a complete classification of the three damping regimes), the driven oscillator (treated separately in the undamped and damped cases), resonance (amplitude response, peak frequency, width, and quality factor), coupled oscillators (matrix form, normal modes, decoupling by a change of coordinates), and the forced coupled oscillator (mode-by-mode response and the antiresonance phenomenon). The treatment then extends to the N -mass chain — built up explicitly from the three-, four-, and five-mass cases — whose normal modes and dispersion relation follow from a spatial exponential ansatz, and closes with the continuum limit, in which the chain becomes a stretched rope and the equations of motion become the wave equation. The general solution of the wave equation leads naturally to the Fourier expansion of initial data, and a closing outlook introduces the wave equation in two and three dimensions. The treatment is entirely Newtonian.

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1 Conventions and Notation

Throughout, $x(t)$ denotes a real displacement from equilibrium, dots denote time derivatives, and i is the imaginary unit. Whether a symbol is real or complex is visible from its mark:

Mark	Meaning	Examples
plain italic	real scalar	$x(t)$, A , ϕ , ω_0
tilde $\sim$	complex-valued <i>function</i> of t (stand-in for a real one)	$\tilde{z}(t)$, $\tilde{F}(t)$
hat $\hat{}$	complex <i>constant</i> (amplitude/phaser)	$\hat{z}$, $\hat{c}_1$, $\hat{D}(\omega)$
arrow $\vec{}$	real column vector	$\vec{x}$, $\vec{v}$
sans-serif	matrix	$\mathbf{A}$

Complex conjugation is written with a superscript star, $\hat{c}^*$ (never a bar, to avoid stacking accents). A hatted quantity may depend on the driving frequency ω — $\hat{D}(\omega)$ is still a “constant” in the relevant sense: constant *in time*. The physical quantity behind any tilde or hat is always its *real part*.

Whenever a parameter is defined for notational convenience, the definition is displayed and its dimensions stated. The recurring definitions are collected here for reference; each is re-introduced where it first appears.

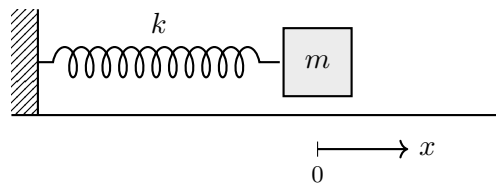
Symbol	Definition	Dimensions	Meaning
ω_0	$\omega_0^2 = k/m$	$[\text{time}]^{-1}$	natural (angular) frequency
β	$2\beta = b/m$	$[\text{time}]^{-1}$	damping constant
ω_d	$\omega_d = \sqrt{\omega_0^2 - \beta^2}$	$[\text{time}]^{-1}$	damped oscillation frequency
f_0	$f_0 = F_0/m$	$[\text{length}][\text{time}]^{-2}$	driving force per unit mass
ω_c	$\omega_c^2 = k_c/m$	$[\text{time}]^{-1}$	coupling frequency (§6)
Q	$Q = \omega_0/(2\beta)$	dimensionless	quality factor
a	lattice spacing	$[\text{length}]$	equilibrium mass separation (§8)
c	$c^2 = T/\mu$	$[\text{length}][\text{time}]^{-1}$	wave speed, $\mu = \text{mass/length}$ (§9)

Convention 1.1 (Complexification). To solve a linear differential equation with a real sinusoidal inhomogeneity $f_0 \cos \omega t$, we will replace it by the complex inhomogeneity $f_0 e^{i\omega t}$, solve for a complex $\tilde{z}(t)$, and recover the physical solution as $x(t) = \text{Re } \tilde{z}(t)$. The justification is Lemma 2.3 below, proved once and used throughout.

2 The Simple Harmonic Oscillator

2.1 Physical setup and equation of motion

A block of mass m slides on a frictionless horizontal surface, attached to a rigid wall by an ideal massless spring of stiffness k . Let x denote the displacement of the block from the position at which the spring is relaxed (the *equilibrium position*); $x > 0$ means the spring is stretched.



By Hooke's law the spring exerts a restoring force $F_{\text{spring}} = -kx$: for $x > 0$ (stretched) the force points in the $-x$ direction, for $x < 0$ (compressed) in the $+x$ direction. Newton's second law, $m\ddot{x} = \sum F$, gives

$$m\ddot{x} = -kx. \quad (1)$$

Convention 2.1 (Natural frequency). Define

$$\boxed{\omega_0^2 \equiv \frac{k}{m}} \quad [\omega_0] = \sqrt{\frac{[\text{force}]/[\text{length}]}{[\text{mass}]} = \frac{1}{[\text{time}]}}. \quad (2)$$

ω_0 is the only parameter of the system with the dimensions of an inverse time; it will turn out to be the angular frequency of the free oscillation, hence the name *natural frequency*.

Dividing (1) by m puts the equation in standard form:

$$\boxed{\ddot{x} + \omega_0^2 x = 0} \quad (3)$$

Remark 2.2 (Why this equation is everywhere). The block and spring stand in for something far more general. Let a particle of mass m move in *any* smooth potential $V(x)$ with a stable equilibrium at x_0 : $V'(x_0) = 0$ and $V''(x_0) > 0$. Taylor expansion about the minimum gives

$$V(x) = V(x_0) + \frac{1}{2} V''(x_0) (x - x_0)^2 + O((x - x_0)^3), \quad (4)$$

the linear term vanishing precisely because x_0 is an equilibrium, so for small displacements Newton's law is Eq. (3) with the effective stiffness $k = V''(x_0)$. Every small oscillation about a stable equilibrium is this equation with its own k — the pendulum, for instance: $V = mg\ell(1 - \cos\theta)$ gives, in the angle coordinate, $\omega_0^2 = g/\ell$. The rest of this document therefore applies wherever equilibria are stable and amplitudes are small, which is why one equation deserves a document.

This is a linear, homogeneous, second-order ordinary differential equation with constant real coefficients. Its solution space is a two-dimensional real vector space: a solution is uniquely determined by the two initial data $x(0)$ and $\dot{x}(0)$,¹ and any linear combination of solutions is again a solution (superposition).

2.2 First solutions: sine and cosine

Equation (3) asks for a function whose second derivative is a negative multiple of itself. Two functions with exactly this property are known from calculus: try $x(t) = \cos \omega_0 t$,

$$\dot{x} = -\omega_0 \sin \omega_0 t, \quad \ddot{x} = -\omega_0^2 \cos \omega_0 t = -\omega_0^2 x \quad \checkmark \quad (5)$$

and $x(t) = \sin \omega_0 t$,

$$\dot{x} = \omega_0 \cos \omega_0 t, \quad \ddot{x} = -\omega_0^2 \sin \omega_0 t = -\omega_0^2 x \quad \checkmark \quad (6)$$

Both solve (3). (The frequency must be ω_0 : trying $\cos \alpha t$ gives $\ddot{x} = -\alpha^2 x$, which matches only for $\alpha^2 = \omega_0^2$.)

¹Uniqueness is the one fact this document imports from the general theory of differential equations, and for the equations used here it has a three-line self-contained proof. Let x_1, x_2 solve $\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = f(t)$ (any $\beta \geq 0$, including the undamped case $\beta = 0$; any driving f) with the same initial data, and set $\delta = x_1 - x_2$. Then δ solves the homogeneous equation with $\delta(0) = \dot{\delta}(0) = 0$, and $E_\delta \equiv \frac{1}{2}\dot{\delta}^2 + \frac{1}{2}\omega_0^2\delta^2$ satisfies $\dot{E}_\delta = \dot{\delta}(\ddot{\delta} + \omega_0^2\delta) = -2\beta\dot{\delta}^2 \leq 0$. Since $E_\delta(0) = 0$ and $E_\delta \geq 0$, E_δ vanishes identically, so $\delta \equiv 0$. The same argument, run with the potential-energy quadratic form of §6, covers the coupled systems. Existence is exhibited constructively throughout, by writing the solutions down.

Linear combinations also work. Because the equation is linear and homogeneous, any combination $x = B_1 \cos \omega_0 t + B_2 \sin \omega_0 t$ with constants B_1, B_2 is again a solution — verify directly:

$$\ddot{x} + \omega_0^2 x = B_1 \underbrace{(-\omega_0^2 \cos \omega_0 t + \omega_0^2 \cos \omega_0 t)}_{=0} + B_2 \underbrace{(-\omega_0^2 \sin \omega_0 t + \omega_0^2 \sin \omega_0 t)}_{=0} = 0. \quad (7)$$

Moreover this two-parameter family is *all* the solutions: the solution space is two-dimensional (a solution is pinned down by the two numbers $x(0), \dot{x}(0)$), and $\cos \omega_0 t, \sin \omega_0 t$ are linearly independent, so they span it. The general real solution is

$$\boxed{x(t) = B_1 \cos \omega_0 t + B_2 \sin \omega_0 t} \quad (8)$$

Initial-value problem. Given $x(0) = x_0$ and $\dot{x}(0) = v_0$:

$$x(0) = B_1 = x_0, \quad (9)$$

$$\dot{x}(t) = -B_1 \omega_0 \sin \omega_0 t + B_2 \omega_0 \cos \omega_0 t \implies \dot{x}(0) = B_2 \omega_0 = v_0 \implies B_2 = \frac{v_0}{\omega_0}. \quad (10)$$

Hence

$$\boxed{x(t) = x_0 \cos \omega_0 t + \frac{v_0}{\omega_0} \sin \omega_0 t} \quad (11)$$

2.3 Amplitude–phase form

The two-term form (8) hides the physics: the motion is in fact a *single* sinusoid. To see it, try to write

$$B_1 \cos \omega_0 t + B_2 \sin \omega_0 t \stackrel{?}{=} A \cos(\omega_0 t - \phi) \quad (12)$$

for some amplitude $A \geq 0$ and phase ϕ . Expand the right side with the cosine addition formula:

$$A \cos(\omega_0 t - \phi) = A \cos \phi \cos \omega_0 t + A \sin \phi \sin \omega_0 t. \quad (13)$$

Since $\cos \omega_0 t$ and $\sin \omega_0 t$ are linearly independent, the two sides agree for all t if and only if the coefficients match:

$$B_1 = A \cos \phi, \quad B_2 = A \sin \phi. \quad (14)$$

These are precisely polar coordinates for the point (B_1, B_2) in the plane: they can always be solved, with

$$\boxed{A = \sqrt{B_1^2 + B_2^2}, \quad \tan \phi = \frac{B_2}{B_1}} \quad (15)$$

(quadrant of ϕ chosen so that $\cos \phi$ has the sign of B_1 and $\sin \phi$ the sign of B_2). The general solution is therefore equally

$$\boxed{x(t) = A \cos(\omega_0 t - \phi)} \quad (16)$$

a single oscillation of angular frequency ω_0 , amplitude A , phase ϕ ; the phase records *when* the maxima occur (at $\omega_0 t = \phi + 2\pi n$), the amplitude *how large* they are. In terms of initial data, from (11):

$$A = \sqrt{x_0^2 + \frac{v_0^2}{\omega_0^2}}, \quad \tan \phi = \frac{v_0}{\omega_0 x_0}. \quad (17)$$

Everything above was elementary: one good guess plus trigonometry. It worked because for *this* equation the correct guess is common knowledge. Add a damping term $2\beta\dot{x}$ (as we will in

§3) and pure sines and cosines no longer solve the equation — the guess would have to be revised to decaying oscillations, with the decay rate and frequency both unknown, and the trigonometric bookkeeping grows accordingly. Before facing that, we set up a method that does not depend on guessing correctly.

2.4 The complex-exponential method

The ansatz. Rather than guessing oscillations, guess the one function family that differentiation cannot change: exponentials,

$$x(t) = e^{\lambda t} \implies \dot{x} = \lambda e^{\lambda t}, \quad \ddot{x} = \lambda^2 e^{\lambda t}, \quad (18)$$

with the constant λ to be determined *by the equation*, not by us. Substituting into (3):

$$\lambda^2 e^{\lambda t} + \omega_0^2 e^{\lambda t} = (\lambda^2 + \omega_0^2) e^{\lambda t} = 0. \quad (19)$$

Since $e^{\lambda t} \neq 0$ for all t , the ansatz solves the equation if and only if λ satisfies the *characteristic equation*

$$\lambda^2 + \omega_0^2 = 0. \quad (20)$$

This has *no real solutions* — consistent with §2.2, where the real solutions turned out to be oscillations, not real exponentials. Here is the decision point: **if we allow the solution to take complex values**, the characteristic equation is solvable,

$$\lambda = \pm i\omega_0, \quad (21)$$

and we obtain two complex solutions $e^{i\omega_0 t}$ and $e^{-i\omega_0 t}$. At the end we will extract the physical (real) part. The next two ingredients make this procedure precise.

Euler's formula. For real θ ,

$$e^{i\theta} = \cos \theta + i \sin \theta. \quad (22)$$

(One way to see this: the power series of $e^{i\theta}$, split into even and odd powers of θ , reproduces the series of $\cos \theta$ and $i \sin \theta$ respectively.) In particular $e^{i\theta}$ traces the unit circle in the complex plane with angular speed 1 in θ , and

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}. \quad (23)$$

So the complex solutions just found are familiar: $e^{\pm i\omega_0 t} = \cos \omega_0 t \pm i \sin \omega_0 t$ — the sine and cosine of §2.2, packaged in complex pairs.

Why taking the real part is legitimate.

Lemma 2.3 (Real and imaginary parts of complex solutions). *Let L be a linear differential operator with real coefficients, e.g. $L = \frac{d^2}{dt^2} + 2\beta \frac{d}{dt} + \omega_0^2$ with $\beta, \omega_0 \in \mathbb{R}$.*

(i) *If a complex-valued function $\tilde{z}(t)$ satisfies $L[\tilde{z}] = 0$, then $\operatorname{Re} \tilde{z}$ and $\operatorname{Im} \tilde{z}$ are real solutions of $L[x] = 0$.*

(ii) *If $\tilde{z}(t)$ satisfies $L[\tilde{z}] = \tilde{g}(t)$ for a complex inhomogeneity $\tilde{g}$, then $\operatorname{Re} \tilde{z}$ satisfies $L[x] = \operatorname{Re} \tilde{g}$.*

Proof. Write $\tilde{z} = u + iv$ with u, v real. Because differentiation acts on real and imaginary parts separately and the coefficients of L are real,

$$L[\tilde{z}] = L[u + iv] = L[u] + iL[v], \quad (24)$$

where $L[u]$ and $L[v]$ are real-valued. (i) If $L[\tilde{z}] = 0$, then both the real part $L[u]$ and the imaginary part $L[v]$ vanish separately, so $u = \operatorname{Re} \tilde{z}$ and $v = \operatorname{Im} \tilde{z}$ solve the homogeneous equation. (ii) If $L[\tilde{z}] = \tilde{g}$, then matching real parts of $L[u] + iL[v] = \operatorname{Re} \tilde{g} + i \operatorname{Im} \tilde{g}$ gives $L[u] = \operatorname{Re} \tilde{g}$. $\square$

Equivalently: since the coefficients are real, taking the complex conjugate of $L[\tilde{z}] = \tilde{g}$ gives $L[\tilde{z}^*] = \tilde{g}^*$, so $\tilde{z}^*$ is a solution of the conjugated problem, and $\operatorname{Re} \tilde{z} = \frac{1}{2}(\tilde{z} + \tilde{z}^*)$ is a solution of the problem with inhomogeneity $\frac{1}{2}(\tilde{g} + \tilde{g}^*) = \operatorname{Re} \tilde{g}$ by linearity. We will invoke Lemma 2.3 every time we complexify.

As a first check, apply part (i) to $\tilde{z} = e^{i\omega_0 t}$: $\operatorname{Re} \tilde{z} = \cos \omega_0 t$ and $\operatorname{Im} \tilde{z} = \sin \omega_0 t$ must be real solutions — and they are exactly the two solutions of §2.2. The complex method has *re-derived* the guess.

General solution and the reality condition. The general complex solution of (3) is the two-parameter family

$$\tilde{z}(t) = \hat{c}_1 e^{i\omega_0 t} + \hat{c}_2 e^{-i\omega_0 t}, \quad \hat{c}_1, \hat{c}_2 \in \mathbb{C}. \quad (25)$$

We want the real solutions among these. Imposing $\tilde{z}(t) = \tilde{z}(t)^*$ for all t :

$$\hat{c}_1 e^{i\omega_0 t} + \hat{c}_2 e^{-i\omega_0 t} = \hat{c}_1^* e^{-i\omega_0 t} + \hat{c}_2^* e^{i\omega_0 t} \iff (\hat{c}_1 - \hat{c}_2^*) e^{i\omega_0 t} = (\hat{c}_1^* - \hat{c}_2) e^{-i\omega_0 t}. \quad (26)$$

Since $e^{i\omega_0 t}$ and $e^{-i\omega_0 t}$ are linearly independent functions, both coefficients must vanish, i.e.

$$\boxed{\hat{c}_2 = \hat{c}_1^*} \quad (\text{reality condition}). \quad (27)$$

Write the one remaining free complex constant in polar form, $\hat{c}_1 = \frac{A}{2} e^{-i\phi}$ with $A \geq 0$ and $\phi \in \mathbb{R}$. Then

$$\begin{aligned} x(t) &= \frac{A}{2} e^{-i\phi} e^{i\omega_0 t} + \frac{A}{2} e^{+i\phi} e^{-i\omega_0 t} = \frac{A}{2} \left(e^{i(\omega_0 t - \phi)} + e^{-i(\omega_0 t - \phi)} \right) \\ &= A \cos(\omega_0 t - \phi), \end{aligned} \quad (28)$$

using (23): the amplitude–phase form (16) of §2.3 drops out *directly* — no trigonometric identity was needed, because the polar decomposition of the complex constant $\hat{c}_1$ is the amplitude–phase decomposition. Equivalently, one may skip the reality condition and simply take

$$x(t) = \operatorname{Re}[\hat{C} e^{i\omega_0 t}], \quad \hat{C} = A e^{-i\phi} \in \mathbb{C}, \quad (29)$$

which is legitimate by Lemma 2.3(i) and contains the same two real parameters packaged into one complex number.

Three equivalent forms. Collecting the results of §2.2–§2.4, the general solution now has three parametrizations,

$$x(t) = \underbrace{B_1 \cos \omega_0 t + B_2 \sin \omega_0 t}_{\text{(I), §2.2}} = \underbrace{A \cos(\omega_0 t - \phi)}_{\text{(II), §2.3}} = \underbrace{\operatorname{Re}[\hat{C} e^{i\omega_0 t}]}_{\text{(III)}} \quad (30)$$

with conversion dictionary (the $B \leftrightarrow (A, \phi)$ part is (14)):

$$B_1 = A \cos \phi, \quad B_2 = A \sin \phi; \quad A = \sqrt{B_1^2 + B_2^2}, \quad \tan \phi = \frac{B_2}{B_1}; \quad \hat{C} = B_1 - iB_2 = Ae^{-i\phi}. \quad (31)$$

(To verify the last identity: $\text{Re}[(B_1 - iB_2)(\cos \omega_0 t + i \sin \omega_0 t)] = B_1 \cos \omega_0 t + B_2 \sin \omega_0 t$.) Form (I) is best for initial conditions, (II) for reading off the physics, (III) for calculation — one complex number carries amplitude as its modulus and phase as (minus) its argument. This packaging is the essential convenience of the complex method, and from the next section on it is the form we compute in.

2.5 Energy

The total mechanical energy is kinetic plus spring potential:

$$E(t) = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2. \quad (32)$$

Differentiate along a solution and use (1):

$$\frac{dE}{dt} = m\dot{x}\ddot{x} + kx\dot{x} = \dot{x}(m\ddot{x} + kx) = \dot{x} \cdot 0 = 0. \quad (33)$$

Energy is conserved. Its value in terms of the amplitude: insert $x = A \cos(\omega_0 t - \phi)$, $\dot{x} = -A\omega_0 \sin(\omega_0 t - \phi)$,

$$\begin{aligned} E &= \frac{1}{2}mA^2\omega_0^2 \sin^2(\omega_0 t - \phi) + \frac{1}{2}kA^2 \cos^2(\omega_0 t - \phi) \\ &= \frac{1}{2}kA^2 [\sin^2(\omega_0 t - \phi) + \cos^2(\omega_0 t - \phi)] = \frac{1}{2}kA^2, \end{aligned} \quad (34)$$

where $m\omega_0^2 = k$ was used to combine the two terms. The energy is set entirely by the amplitude, and oscillates back and forth between kinetic and potential form at angular frequency $2\omega_0$.

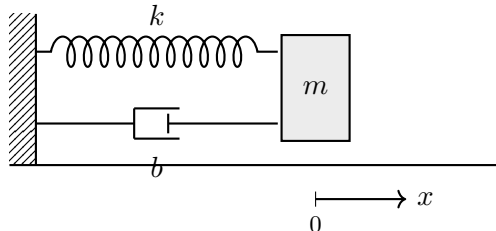
3 The Damped Harmonic Oscillator

3.1 Physical setup and equation of motion

Add to the previous system a linear drag force

$$F_{\text{drag}} = -b\dot{x}, \quad b > 0, \quad (35)$$

opposing the velocity. (Physically: viscous friction, valid as the leading-order model at low speeds; a dashpot in the mechanical idealization.)



Newton's second law now reads $m\ddot{x} = -kx - b\dot{x}$, i.e.

$$m\ddot{x} + b\dot{x} + kx = 0. \quad (36)$$

Convention 3.1 (Damping constant). Define

$$\boxed{2\beta \equiv \frac{b}{m}} \quad [\beta] = \frac{1}{[\text{time}]}. \quad (37)$$

The factor of 2 is chosen purely so that the roots of the characteristic equation below come out without factors of $\frac{1}{2}$.

Dividing by m , the standard form is

$$\boxed{\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = 0} \quad (38)$$

3.2 Characteristic equation and the discriminant

The same exponential ansatz $x = e^{\lambda t}$ applies verbatim — this is the payoff of having set up the method in §2.4: nothing new is needed. Substituting,

$$(\lambda^2 + 2\beta\lambda + \omega_0^2)e^{\lambda t} = 0 \quad \Longleftrightarrow \quad \lambda^2 + 2\beta\lambda + \omega_0^2 = 0, \quad (39)$$

with roots (quadratic formula)

$$\boxed{\lambda_{\pm} = -\beta \pm \sqrt{\beta^2 - \omega_0^2}} \quad (40)$$

The qualitative behavior is decided by the sign of the discriminant $\beta^2 - \omega_0^2$, giving *three cases*:

Case	Condition	Roots
underdamped	$\beta < \omega_0$	complex conjugate pair, $\lambda_{\pm} = -\beta \pm i\omega_d$
critically damped	$\beta = \omega_0$	real double root, $\lambda = -\beta$
overdamped	$\beta > \omega_0$	two distinct negative real roots

3.3 Case I: underdamped ($\beta < \omega_0$)

The discriminant is negative; write

$$\sqrt{\beta^2 - \omega_0^2} = i\sqrt{\omega_0^2 - \beta^2} \quad \text{and define} \quad \boxed{\omega_d \equiv \sqrt{\omega_0^2 - \beta^2}} \quad (41)$$

(the *damped frequency*; note $0 < \omega_d < \omega_0$). Then $\lambda_{\pm} = -\beta \pm i\omega_d$ and the general complex solution is

$$\tilde{z}(t) = \hat{c}_1 e^{(-\beta + i\omega_d)t} + \hat{c}_2 e^{(-\beta - i\omega_d)t} = e^{-\beta t} \left(\hat{c}_1 e^{i\omega_d t} + \hat{c}_2 e^{-i\omega_d t} \right). \quad (42)$$

The bracket is exactly the SHO structure of (25) with $\omega_0 \rightarrow \omega_d$, so the reality condition and polar parametrization go through unchanged ($\hat{c}_2 = \hat{c}_1^*$, $\hat{c}_1 = \frac{A}{2}e^{-i\phi}$):

$$\boxed{x(t) = A e^{-\beta t} \cos(\omega_d t - \phi)} \quad (43)$$

An oscillation at the *reduced* frequency ω_d , inside an exponentially decaying envelope $\pm A e^{-\beta t}$. Two remarks:

- The zero crossings are still equally spaced (period $T_d = 2\pi/\omega_d$), but $T_d > 2\pi/\omega_0$: damping slows the oscillation.
- Over one period the amplitude shrinks by the fixed factor $e^{-\beta T_d}$; the *logarithmic decrement* $\Lambda \equiv \beta T_d = 2\pi\beta/\omega_d$ is a standard experimental measure of damping — for weak damping $\Lambda \approx \pi/Q$, tying this time-domain measure to the linewidth measure of §5.3.

Initial-value problem. With $x(0) = x_0$, $\dot{x}(0) = v_0$, it is easiest to use the form $x = e^{-\beta t}(B_1 \cos \omega_d t + B_2 \sin \omega_d t)$:

$$x(0) = B_1 = x_0, \quad (44)$$

$$\begin{aligned} \dot{x}(t) &= -\beta e^{-\beta t}(B_1 \cos \omega_d t + B_2 \sin \omega_d t) + e^{-\beta t}(-B_1 \omega_d \sin \omega_d t + B_2 \omega_d \cos \omega_d t), \\ \dot{x}(0) &= -\beta B_1 + \omega_d B_2 = v_0 \implies B_2 = \frac{v_0 + \beta x_0}{\omega_d}. \end{aligned} \quad (45)$$

$$\boxed{x(t) = e^{-\beta t} \left[x_0 \cos \omega_d t + \frac{v_0 + \beta x_0}{\omega_d} \sin \omega_d t \right]} \quad (46)$$

3.4 Case II: overdamped ($\beta > \omega_0$)

Both roots are real; write $\gamma \equiv \sqrt{\beta^2 - \omega_0^2} > 0$, so

$$\lambda_{\pm} = -\beta \pm \gamma, \quad \lambda_- < -\beta < \lambda_+ < 0. \quad (47)$$

(That $\lambda_+ < 0$: $\gamma = \sqrt{\beta^2 - \omega_0^2} < \beta$ because $\omega_0^2 > 0$.) The general solution is a sum of two pure decays, no oscillation:

$$\boxed{x(t) = C_+ e^{\lambda_+ t} + C_- e^{\lambda_- t}} \quad (48)$$

Imposing $x(0) = x_0$, $\dot{x}(0) = v_0$ gives the linear system

$$C_+ + C_- = x_0, \quad \lambda_+ C_+ + \lambda_- C_- = v_0. \quad (49)$$

Solving (multiply the first equation by λ_- and subtract, then by λ_+ and subtract):

$$C_+ = \frac{v_0 - \lambda_- x_0}{\lambda_+ - \lambda_-}, \quad C_- = \frac{\lambda_+ x_0 - v_0}{\lambda_+ - \lambda_-}. \quad (50)$$

For example, released from rest ($v_0 = 0$),

$$x(t) = \frac{\lambda_+ e^{\lambda_- t} - \lambda_- e^{\lambda_+ t}}{\lambda_+ - \lambda_-} x_0, \quad (51)$$

which one verifies directly: at $t = 0$ the numerator is $\lambda_+ - \lambda_-$, giving $x(0) = x_0$; the derivative's numerator at $t = 0$ is $\lambda_+ \lambda_- - \lambda_- \lambda_+ = 0$, giving $\dot{x}(0) = 0$.

Remark 3.2 (Long-time behavior). Since $|\lambda_+| < |\lambda_-|$, the $e^{\lambda_+ t}$ term decays more slowly and dominates at late times: the approach to equilibrium is governed by the *smaller* decay rate $|\lambda_+| = \beta - \gamma$. Increasing the damping β beyond ω_0 actually makes the late-time return to equilibrium *slower*: as $\beta \rightarrow \infty$, $\beta - \sqrt{\beta^2 - \omega_0^2} = \frac{\omega_0^2}{\beta + \sqrt{\beta^2 - \omega_0^2}} \rightarrow \frac{\omega_0^2}{2\beta} \rightarrow 0$.

Remark 3.3 (At most one zero crossing). $x(t) = 0$ requires $C_+ e^{\lambda_+ t} = -C_- e^{\lambda_- t}$, i.e. $e^{(\lambda_+ - \lambda_-)t} = -C_-/C_+$. The left side is strictly monotone in t , so there is at most one solution: an overdamped oscillator can pass through equilibrium at most once.

3.5 Case III: critically damped ($\beta = \omega_0$)

The characteristic equation has the *double* root $\lambda = -\beta$, so the ansatz produces only one solution, $e^{-\beta t}$. The solution space is two-dimensional, so a second, linearly independent solution must exist; we find it by *reduction of order*. Substitute

$$x(t) = u(t) e^{-\beta t} \quad (52)$$

with u to be determined. Compute the derivatives:

$$\dot{x} = \dot{u} e^{-\beta t} - \beta u e^{-\beta t} = (\dot{u} - \beta u) e^{-\beta t}, \quad (53)$$

$$\ddot{x} = (\ddot{u} - \beta \dot{u}) e^{-\beta t} - \beta(\dot{u} - \beta u) e^{-\beta t} = (\ddot{u} - 2\beta \dot{u} + \beta^2 u) e^{-\beta t}. \quad (54)$$

Substitute into (38) with $\omega_0^2 = \beta^2$:

$$\begin{aligned} 0 &= \ddot{x} + 2\beta \dot{x} + \beta^2 x \\ &= \left[(\ddot{u} - 2\beta \dot{u} + \beta^2 u) + 2\beta(\dot{u} - \beta u) + \beta^2 u \right] e^{-\beta t} \\ &= \left[\ddot{u} + (-2\beta + 2\beta)\dot{u} + (\beta^2 - 2\beta^2 + \beta^2)u \right] e^{-\beta t} = \ddot{u} e^{-\beta t}. \end{aligned} \quad (55)$$

Every term in u and $\dot{u}$ cancels *exactly because* $\lambda = -\beta$ *is a double root*, leaving simply

$$\ddot{u} = 0 \implies u(t) = C_1 + C_2 t. \quad (56)$$

Hence the general solution in the critical case is

$$\boxed{x(t) = (C_1 + C_2 t) e^{-\beta t}} \quad (57)$$

The second solution $t e^{-\beta t}$ is not an exponential — it is the degenerate limit in which the two exponentials $e^{\lambda_{\pm} t}$ merge. (Indeed $\lim_{\gamma \rightarrow 0} \frac{e^{(-\beta+\gamma)t} - e^{(-\beta-\gamma)t}}{2\gamma} = t e^{-\beta t}$, a difference quotient in the root itself.)

Initial conditions: $x(0) = C_1 = x_0$ and $\dot{x}(0) = C_2 - \beta C_1 = v_0 \Rightarrow C_2 = v_0 + \beta x_0$:

$$x(t) = [x_0 + (v_0 + \beta x_0) t] e^{-\beta t}. \quad (58)$$

Remark 3.4 (Fastest return). Among the three regimes at fixed ω_0 , compare the decay rate of the slowest-decaying term in each solution: β for the underdamped envelope, $\beta - \gamma$ for the overdamped case, and ω_0 (up to the polynomial factor t) for critical damping. Since $\beta < \omega_0$ in the first case and $\beta - \gamma < \omega_0$ in the second (both strict), the critical rate is the largest — and critical damping is the only regime that combines this fastest decay with no zero crossings. This is why measuring instruments and door closers are tuned near critical damping.

3.6 Energy decay and the quality factor

With drag, energy is no longer conserved. Differentiating $E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2$ along a solution of $m\ddot{x} = -kx - b\dot{x}$:

$$\frac{dE}{dt} = \dot{x}(m\ddot{x} + kx) = \dot{x}(-b\dot{x}) = -b\dot{x}^2 \leq 0. \quad (59)$$

Energy decreases monotonically (except at the isolated instants where $\dot{x} = 0$), at exactly the rate at which the drag force does negative work.

For weak damping ($\beta \ll \omega_0$), the envelope of (43) gives $x_{\max} \propto e^{-\beta t}$, and since $E \propto A^2$,

$$E(t) \approx E(0) e^{-2\beta t}. \quad (60)$$

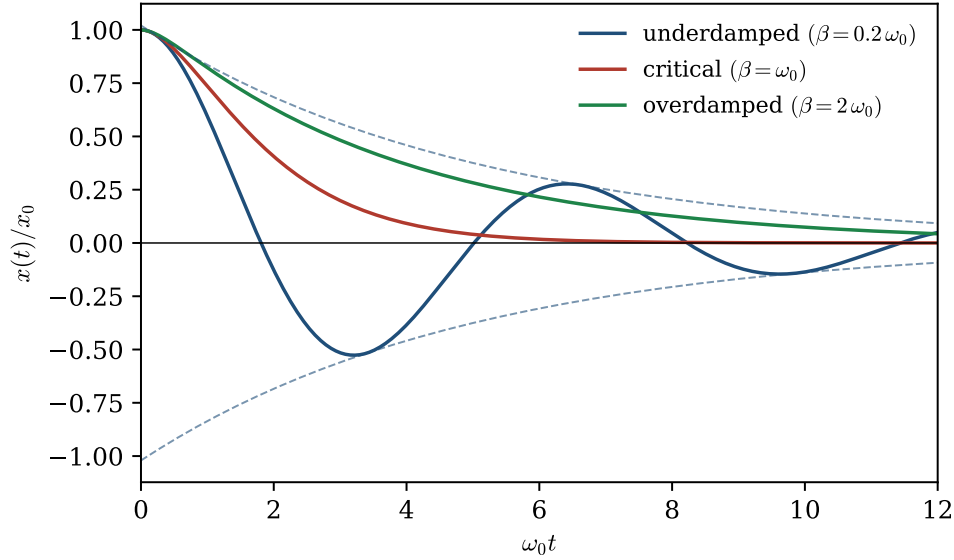


Figure 1: The three damping regimes for the same initial condition $x(0) = x_0$, $\dot{x}(0) = 0$. Underdamped motion (with its envelope $\pm \text{const} e^{-\beta t}$, dashed), critical damping, and overdamped motion.

Convention 3.5 (Quality factor).

$$Q \equiv \frac{\omega_0}{2\beta} \quad (61)$$

Interpretation: the energy decays by e^{-1} in a time $1/(2\beta)$, during which the oscillator completes $\frac{\omega_0}{2\pi} \cdot \frac{1}{2\beta} = \frac{Q}{2\pi}$ cycles. Large Q = many oscillations before the energy is dissipated. The same Q will reappear in §5 as the sharpness of the resonance peak — a nontrivial consistency check.

4 The Forced Harmonic Oscillator

We now apply an external driving force $F(t) = F_0 \cos \omega t$ to the mass. Note carefully: ω is the *driving* frequency, an external knob we control; it is distinct from the system parameters ω_0 and β . Define the force per unit mass

$$f_0 \equiv \frac{F_0}{m} \quad [f_0] = \frac{[\text{length}]}{[\text{time}]^2}. \quad (62)$$

We treat the two cases separately, in increasing order of realism:

§4.1 **Case A: undamped, driven** — cleanest formulas; exhibits beats and the resonance catastrophe.

§4.2 **Case B: damped, driven** — the physically complete case; exhibits transients, steady state, and finite resonance.

Structure of the solution space

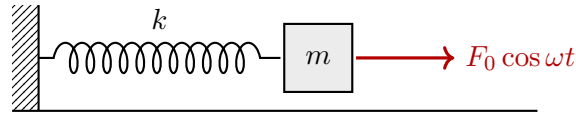
Both cases are *inhomogeneous* linear equations, $L[x] = f_0 \cos \omega t$. If x_p is any one (*particular*) solution and x_h is the general solution of the homogeneous equation $L[x] = 0$, then $x_p + x_h$ solves

the inhomogeneous equation, and conversely any two solutions differ by a homogeneous solution ($L[x_1 - x_2] = f - f = 0$). Hence

$$\boxed{\text{general solution} = \text{particular solution} + \text{general homogeneous solution}} \quad (63)$$

The homogeneous solutions are exactly those found in the previous two sections. The new task in this section is to construct one particular solution — and this is where the complex method of Convention 1.1 shows its value.

4.1 Case A: undamped and driven



The equation of motion is

$$\boxed{\ddot{x} + \omega_0^2 x = f_0 \cos \omega t} \quad (64)$$

Particular solution by complexification. Following Convention 1.1, consider the complexified problem

$$\ddot{\tilde{z}} + \omega_0^2 \tilde{z} = f_0 e^{i\omega t}, \quad (65)$$

whose right-hand side has real part $f_0 \cos \omega t$; by Lemma 2.3(ii), $x_p = \text{Re } \tilde{z}_p$ will solve (64). Since the right-hand side is an exponential and the left-hand side does not change the form of exponentials, try

$$\tilde{z}_p(t) = \hat{z} e^{i\omega t}, \quad \hat{z} \in \mathbb{C} \text{ constant.} \quad (66)$$

Then $\ddot{\tilde{z}}_p = -\omega^2 \hat{z} e^{i\omega t}$, and substituting into (65):

$$(-\omega^2 + \omega_0^2) \hat{z} e^{i\omega t} = f_0 e^{i\omega t} \implies \hat{z} = \frac{f_0}{\omega_0^2 - \omega^2} \quad (\omega \neq \omega_0). \quad (67)$$

The differential equation has been reduced to a one-line algebraic equation. Here $\hat{z}$ happens to be real, so

$$\boxed{x_p(t) = \frac{f_0}{\omega_0^2 - \omega^2} \cos \omega t \quad (\omega \neq \omega_0)} \quad (68)$$

Observations before adding the homogeneous part:

- The particular solution oscillates at the *driving* frequency ω , not at ω_0 .
- For $\omega < \omega_0$ the response is *in phase* with the drive (positive coefficient); for $\omega > \omega_0$ it is *exactly out of phase* (negative coefficient, i.e. phase lag π). The switch is abrupt — damping will smooth it out in Case B.
- The amplitude blows up as $\omega \rightarrow \omega_0$. The excluded case $\omega = \omega_0$ is treated below.

4.1.1 Sub-case A1: off resonance ($\omega \neq \omega_0$), beats

By (63) the general solution is

$$x(t) = \frac{f_0}{\omega_0^2 - \omega^2} \cos \omega t + B_1 \cos \omega_0 t + B_2 \sin \omega_0 t. \quad (69)$$

Take the oscillator to start *from rest at equilibrium*: $x(0) = 0$, $\dot{x}(0) = 0$. Then

$$x(0) = \frac{f_0}{\omega_0^2 - \omega^2} + B_1 = 0 \quad \implies \quad B_1 = -\frac{f_0}{\omega_0^2 - \omega^2}, \quad (70)$$

$$\dot{x}(0) = 0 + B_2 \omega_0 = 0 \quad \implies \quad B_2 = 0, \quad (71)$$

so

$$x(t) = \frac{f_0}{\omega_0^2 - \omega^2} (\cos \omega t - \cos \omega_0 t). \quad (72)$$

To expose the structure, use the product identity. From the addition formulas,

$$\cos(P + Q) = \cos P \cos Q - \sin P \sin Q, \quad (73)$$

$$\cos(P - Q) = \cos P \cos Q + \sin P \sin Q, \quad (74)$$

subtracting gives $\cos(P - Q) - \cos(P + Q) = 2 \sin P \sin Q$. Setting $P = \frac{(\omega_0 + \omega)t}{2}$, $Q = \frac{(\omega_0 - \omega)t}{2}$, so that $P - Q = \omega t$ and $P + Q = \omega_0 t$:

$$\cos \omega t - \cos \omega_0 t = 2 \sin\left(\frac{(\omega_0 + \omega)t}{2}\right) \sin\left(\frac{(\omega_0 - \omega)t}{2}\right). \quad (75)$$

Therefore

$$x(t) = \underbrace{\frac{2f_0}{\omega_0^2 - \omega^2} \sin\left(\frac{(\omega_0 - \omega)t}{2}\right)}_{\text{slow envelope}} \cdot \underbrace{\sin\left(\frac{(\omega_0 + \omega)t}{2}\right)}_{\text{fast oscillation}} \quad (76)$$

When the drive is close to resonance, $\omega \approx \omega_0$, the first factor oscillates slowly (frequency $\frac{|\omega_0 - \omega|}{2}$) and acts as a modulating envelope on the fast oscillation (frequency $\approx \omega_0$): the amplitude repeatedly builds up and dies back down. These are *beats* (Fig. 2). The maximum displacement, $\left|\frac{2f_0}{\omega_0^2 - \omega^2}\right|$, grows as $\omega \rightarrow \omega_0$ — the beats become taller and slower, foreshadowing the resonant case.

4.1.2 Sub-case A2: on resonance ($\omega = \omega_0$)

At $\omega = \omega_0$ the ansatz $\hat{z} e^{i\omega_0 t}$ fails — $e^{i\omega_0 t}$ already solves the homogeneous equation, so the left-hand side annihilates it and no constant $\hat{z}$ can produce the required right-hand side. We give two independent derivations of the correct particular solution.

Derivation 1: limit of the off-resonance solution. Fix t and let $\omega \rightarrow \omega_0$ in (72). Both numerator and denominator vanish, so apply l'Hôpital's rule *in the variable ω* :

$$\lim_{\omega \rightarrow \omega_0} \frac{f_0 (\cos \omega t - \cos \omega_0 t)}{\omega_0^2 - \omega^2} = \lim_{\omega \rightarrow \omega_0} \frac{f_0 \cdot (-t \sin \omega t)}{-2\omega} = \frac{f_0 t \sin \omega_0 t}{2\omega_0}. \quad (77)$$

So the from-rest resonant solution is

$$x(t) = \frac{f_0}{2\omega_0} t \sin \omega_0 t \quad (78)$$

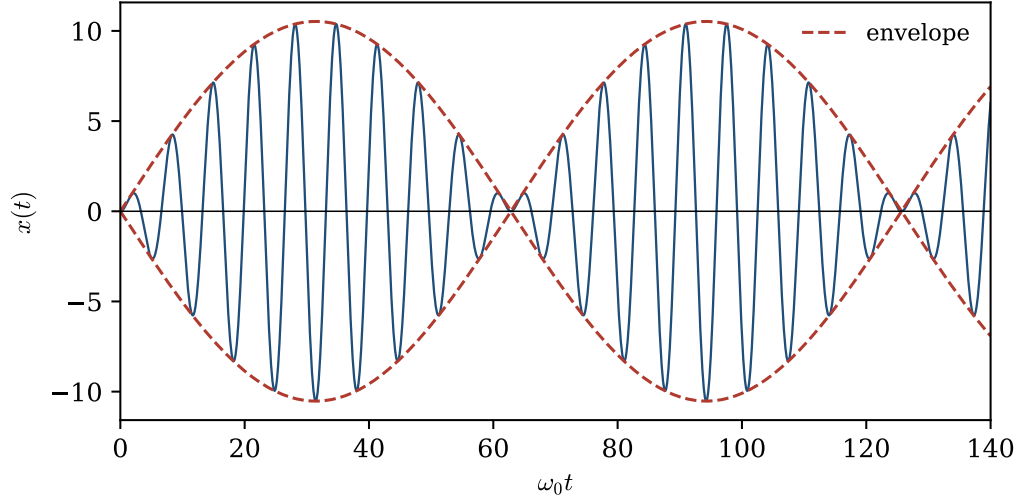


Figure 2: Undamped driven oscillator started from rest, $\omega = 0.9\omega_0$: beats. Dashed: the slow envelope from Eq. (76).

Derivation 2: direct complex ansatz. Guided by the critical-damping experience of §3.5 (when a root is “used up,” multiply by t), try

$$\tilde{z}_p(t) = \hat{c} t e^{i\omega_0 t}, \quad \hat{c} \in \mathbb{C}. \quad (79)$$

Compute the derivatives carefully:

$$\dot{\tilde{z}}_p = \hat{c} e^{i\omega_0 t} + \hat{c} t (i\omega_0) e^{i\omega_0 t} = \hat{c} (1 + i\omega_0 t) e^{i\omega_0 t}, \quad (80)$$

$$\ddot{\tilde{z}}_p = \hat{c} (i\omega_0) e^{i\omega_0 t} + \hat{c} (1 + i\omega_0 t) (i\omega_0) e^{i\omega_0 t} = \hat{c} (2i\omega_0 - \omega_0^2 t) e^{i\omega_0 t}. \quad (81)$$

Substitute into $\ddot{z} + \omega_0^2 z = f_0 e^{i\omega_0 t}$:

$$\hat{c} (2i\omega_0 - \omega_0^2 t) e^{i\omega_0 t} + \omega_0^2 \hat{c} t e^{i\omega_0 t} = \hat{c} \cdot 2i\omega_0 e^{i\omega_0 t} \stackrel{!}{=} f_0 e^{i\omega_0 t}. \quad (82)$$

The t -proportional terms cancel identically (same mechanism as in §3.5), leaving

$$\hat{c} = \frac{f_0}{2i\omega_0} = -\frac{if_0}{2\omega_0}. \quad (83)$$

Extract the real part:

$$x_p = \text{Re} \left[-\frac{if_0}{2\omega_0} t (\cos \omega_0 t + i \sin \omega_0 t) \right] = \frac{f_0}{2\omega_0} t \sin \omega_0 t, \quad (84)$$

in agreement with Derivation 1. (This particular solution happens to satisfy $x_p(0) = \dot{x}_p(0) = 0$, so it *is* the from-rest solution; in general one would still add homogeneous terms.)

The displacement grows *linearly without bound* (Fig. 3): the drive, phase-locked a quarter cycle ahead of the displacement, does positive work on every cycle and there is no dissipation to absorb it. This is the *resonance catastrophe* — the purest form of resonance, visible here without any damping to obscure it. Any real system either has damping (Case B) or leaves the linear regime.

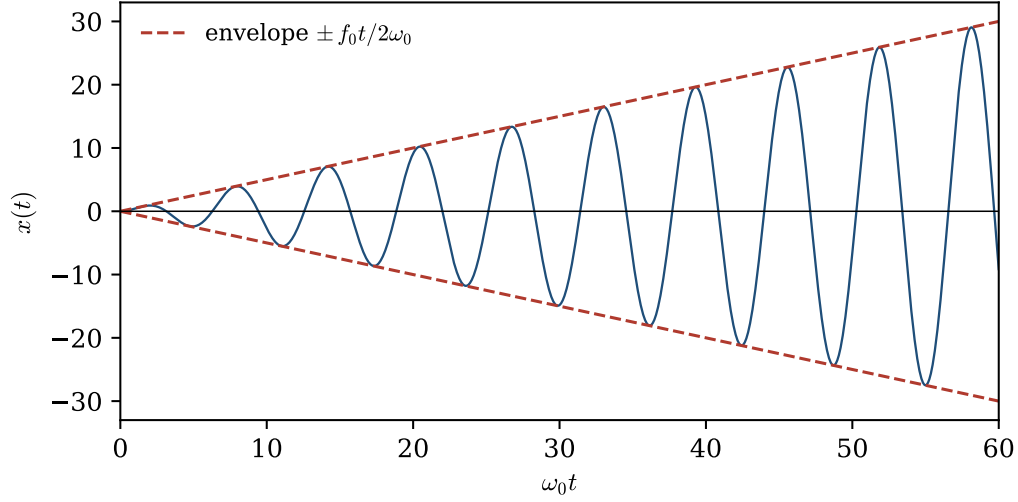


Figure 3: Resonant driving of the undamped oscillator, $\omega = \omega_0$: linear growth $x = \frac{f_0}{2\omega_0} t \sin \omega_0 t$.

4.2 Case B: damped and driven

The full equation of motion:

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = f_0 \cos \omega t \quad (85)$$

Particular solution by complexification. Complexify as before:

$$\ddot{\tilde{z}} + 2\beta\dot{\tilde{z}} + \omega_0^2 \tilde{z} = f_0 e^{i\omega t}, \quad \tilde{z}_p = \hat{z} e^{i\omega t}. \quad (86)$$

Each derivative pulls down a factor of $i\omega$:

$$(-\omega^2 + 2i\beta\omega + \omega_0^2) \hat{z} e^{i\omega t} = f_0 e^{i\omega t} \implies \hat{z} = \frac{f_0}{\omega_0^2 - \omega^2 + 2i\beta\omega} \quad (87)$$

One line. Note that the denominator now has a nonzero imaginary part for all $\omega > 0$, so it never vanishes: with damping there is no excluded frequency, and this single formula covers everything.

Extracting amplitude and phase. Write the denominator in polar form. With

$$\hat{D}(\omega) \equiv (\omega_0^2 - \omega^2) + i(2\beta\omega) = |\hat{D}| e^{i\delta}, \quad (88)$$

the modulus and argument are

$$|\hat{D}(\omega)| = \sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}, \quad \cos \delta = \frac{\omega_0^2 - \omega^2}{|\hat{D}|}, \quad \sin \delta = \frac{2\beta\omega}{|\hat{D}|}. \quad (89)$$

Since $2\beta\omega > 0$, we have $\sin \delta > 0$, so $\delta \in (0, \pi)$: the argument is pinned to the upper half plane, and the often-quoted formula

$$\tan \delta = \frac{2\beta\omega}{\omega_0^2 - \omega^2} \quad (90)$$

must be read with this quadrant convention ($\delta \in (0, \frac{\pi}{2})$ for $\omega < \omega_0$; $\delta \in (\frac{\pi}{2}, \pi)$ for $\omega > \omega_0$; naively applying arctan would wrongly return a negative angle above resonance). Then

$$\hat{z} = \frac{f_0}{|\hat{D}| e^{i\delta}} = \frac{f_0}{|\hat{D}|} e^{-i\delta}, \quad \tilde{z}_p(t) = \frac{f_0}{|\hat{D}|} e^{i(\omega t - \delta)}, \quad (91)$$

and taking the real part (Lemma 2.3(ii)):

$$x_p(t) = A(\omega) \cos(\omega t - \delta(\omega)), \quad A(\omega) = \frac{f_0}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2 \omega^2}} \quad (92)$$

The steady-state response is a sinusoid at the driving frequency, with amplitude $A(\omega)$ and *phase lag* $\delta(\omega)$ behind the drive. The entire frequency response — both numbers — was carried by the single complex quantity $\hat{z}$: modulus = amplitude, minus argument = phase. This is the punchline of the complex method.

Limiting behaviors.

Regime	Amplitude	Phase
$\omega \rightarrow 0$	$A \rightarrow f_0/\omega_0^2 = F_0/k$ (static/stiffness-controlled)	$\delta \rightarrow 0$ (in phase)
$\omega = \omega_0$	$A = \frac{f_0}{2\beta\omega_0}$	$\delta = \frac{\pi}{2}$ (quadrature)
$\omega \rightarrow \infty$	$A \approx f_0/\omega^2 \rightarrow 0$ (mass-controlled)	$\delta \rightarrow \pi$ (antiphase)

At low frequency the mass adiabatically tracks the force ($kx \approx F$); at high frequency the mass cannot keep up and $m\ddot{x} \approx F$ forces the response into antiphase; at ω_0 the response lags by exactly a quarter period regardless of the damping strength.

4.2.1 Contrast: the real undetermined-coefficients route

For comparison, solve the same problem without complex numbers. Since derivatives of $\cos \omega t$ generate $\sin \omega t$ and vice versa, the minimal real ansatz needs both:

$$x_p = a \cos \omega t + b \sin \omega t. \quad (93)$$

Then

$$\dot{x}_p = -a\omega \sin \omega t + b\omega \cos \omega t, \quad (94)$$

$$\ddot{x}_p = -a\omega^2 \cos \omega t - b\omega^2 \sin \omega t. \quad (95)$$

Substitute into (85) and collect the coefficients of $\cos \omega t$ and $\sin \omega t$ separately (they are linearly independent functions, so each must match):

$$\begin{aligned} \cos \omega t : & \quad -a\omega^2 + 2\beta\omega b + \omega_0^2 a = f_0, \\ \sin \omega t : & \quad -b\omega^2 - 2\beta\omega a + \omega_0^2 b = 0, \end{aligned} \quad (96)$$

i.e. the 2×2 linear system

$$\begin{pmatrix} \omega_0^2 - \omega^2 & 2\beta\omega \\ -2\beta\omega & \omega_0^2 - \omega^2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} f_0 \\ 0 \end{pmatrix}. \quad (97)$$

The determinant is $\Delta = (\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2 = |\hat{D}|^2$, and Cramer's rule gives

$$a = \frac{f_0(\omega_0^2 - \omega^2)}{\Delta}, \quad b = \frac{2\beta\omega f_0}{\Delta}. \quad (98)$$

Converting (a, b) to amplitude-phase form via §2.3 reproduces (92) exactly: $A = \sqrt{a^2 + b^2} = f_0\sqrt{\Delta}/\Delta = f_0/|\hat{D}|$ and $\tan \delta = b/a = 2\beta\omega/(\omega_0^2 - \omega^2)$.

Remark 4.1. Both routes are elementary, but note what the real route had to do: double the ansatz, expand derivatives into cross terms, sort two trigonometric families, solve a 2×2 system, and re-assemble amplitude and phase at the end. The complex route did all of this implicitly, because $e^{i\omega t}$ is an eigenfunction of d/dt while $\{\cos, \sin\}$ merely span an invariant 2-dimensional subspace that d/dt mixes. In the coupled systems of §7 the same bookkeeping would multiply per mode; we will use the complex method without further comment.

4.2.2 Transient plus steady state

By (63), the *general* solution of (85) in the underdamped case is

$$x(t) = \underbrace{A \cos(\omega t - \delta)}_{\text{steady state}} + \underbrace{e^{-\beta t}(B_1 \cos \omega_d t + B_2 \sin \omega_d t)}_{\text{transient}}, \quad (99)$$

with A, δ fixed by (92) and B_1, B_2 fixed by the initial conditions. Whatever B_1, B_2 are, the second term dies out on the timescale $1/\beta$:

after a few multiples of $1/\beta$, $x(t) \rightarrow A(\omega) \cos(\omega t - \delta(\omega))$, independent of initial conditions.

(100)

The system forgets how it started; only the drive's imprint survives. This is why $A(\omega)$ and $\delta(\omega)$ are *the* experimentally meaningful response functions — and their reach is wider than a single sinusoid: the equation is linear, so a drive assembled as a sum of sinusoids produces the sum of the corresponding steady states. Knowing $A(\omega)$ and $\delta(\omega)$ for every ω is knowing the long-time response to any drive that can be so decomposed (§9.4 develops the decomposition idea itself, for initial data).

Worked example: starting from rest. Impose $x(0) = 0$, $\dot{x}(0) = 0$ on (99).

$$x(0) = A \cos \delta + B_1 = 0 \implies B_1 = -A \cos \delta. \quad (101)$$

$$\dot{x}(t) = -A\omega \sin(\omega t - \delta) + e^{-\beta t} \left[-\beta(B_1 \cos \omega_d t + B_2 \sin \omega_d t) + \omega_d(-B_1 \sin \omega_d t + B_2 \cos \omega_d t) \right],$$

$$\dot{x}(0) = A\omega \sin \delta - \beta B_1 + \omega_d B_2 = 0 \implies B_2 = -\frac{A(\omega \sin \delta + \beta \cos \delta)}{\omega_d}, \quad (102)$$

where $-\beta B_1 = +\beta A \cos \delta$ was used. The response therefore rings at ω_d while settling into the drive; interference between the two frequencies produces transient beats when $\omega \approx \omega_d$, which then decay — the damped remnant of Fig. 2.

5 Resonance

This section analyzes the steady-state amplitude (92) as a function of the driving frequency:

$$A(\omega) = \frac{f_0}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}}. \quad (103)$$

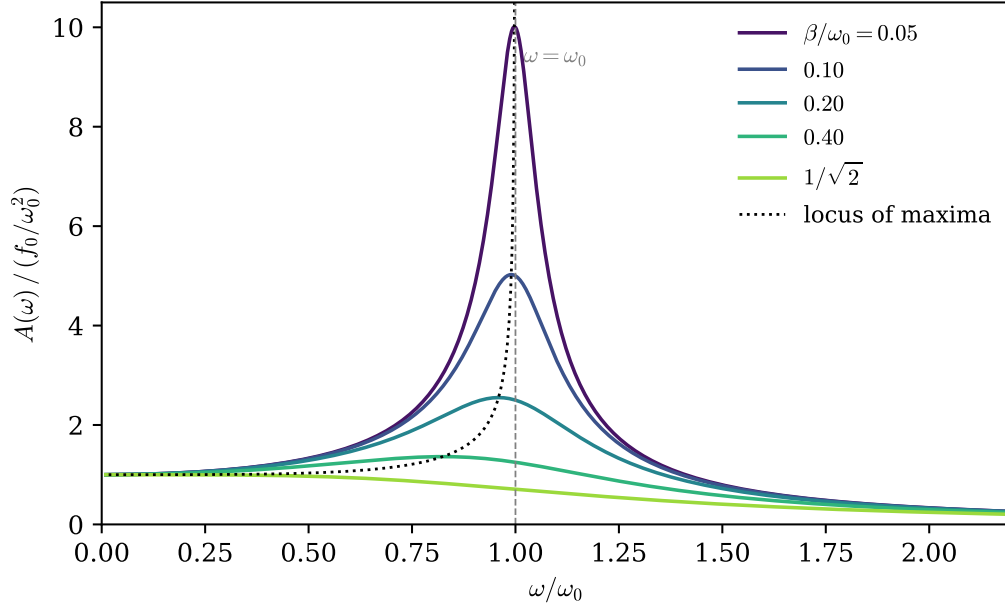


Figure 4: Amplitude response $A(\omega)$ for several damping strengths (amplitude in units of the static response f_0/ω_0^2). The dotted curve is the locus of maxima $(\omega_{\text{peak}}, A_{\text{max}})$ as β varies; the peak moves *left* of ω_0 and disappears at $\beta = \omega_0/\sqrt{2}$.

5.1 Peak frequency and peak amplitude

Maximizing A is equivalent to minimizing the radicand. Substitute $u \equiv \omega^2 \geq 0$ and define

$$g(u) = (\omega_0^2 - u)^2 + 4\beta^2 u. \quad (104)$$

Then

$$g'(u) = -2(\omega_0^2 - u) + 4\beta^2, \quad g''(u) = 2 > 0, \quad (105)$$

so g is a convex parabola in u with unique minimum where $g'(u) = 0$:

$$u_* = \omega_0^2 - 2\beta^2. \quad (106)$$

Two cases:

- If $\beta < \omega_0/\sqrt{2}$, then $u_* > 0$ and the amplitude peaks at

$$\boxed{\omega_{\text{peak}} = \sqrt{\omega_0^2 - 2\beta^2}} \quad (107)$$

- If $\beta \geq \omega_0/\sqrt{2}$, then $u_* \leq 0$: on the physical domain $u \geq 0$ the function g is increasing (since $g'(0) = -2\omega_0^2 + 4\beta^2 \geq 0$), so $A(\omega)$ is monotonically decreasing and there is *no resonance peak at all* — the maximum response is the static one at $\omega = 0$.

Peak height, for $\beta < \omega_0/\sqrt{2}$: substitute u_* back,

$$\begin{aligned} g(u_*) &= (\omega_0^2 - (\omega_0^2 - 2\beta^2))^2 + 4\beta^2(\omega_0^2 - 2\beta^2) \\ &= 4\beta^4 + 4\beta^2\omega_0^2 - 8\beta^4 = 4\beta^2(\omega_0^2 - \beta^2) = 4\beta^2\omega_d^2, \end{aligned} \quad (108)$$

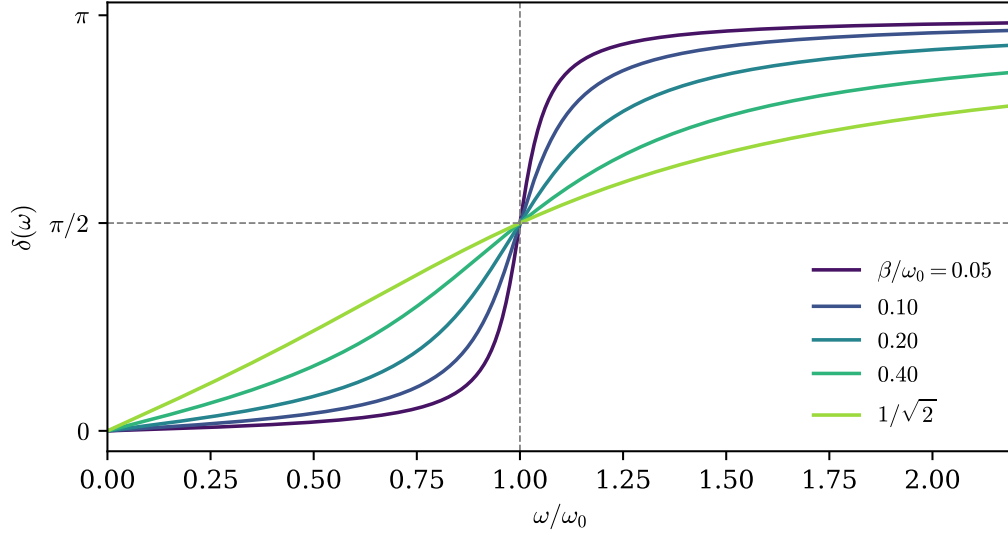


Figure 5: Phase lag $\delta(\omega)$ for the same damping strengths. All curves pass through $\delta = \pi/2$ exactly at $\omega = \omega_0$; the crossing sharpens as $\beta \rightarrow 0$, approaching the undamped step of §4.1.

so

$$A_{\max} = \frac{f_0}{2\beta\omega_d} = \frac{f_0}{2\beta\sqrt{\omega_0^2 - \beta^2}} \quad (109)$$

For weak damping, $A_{\max} \approx \frac{f_0}{2\beta\omega_0} = Q \cdot \frac{f_0}{\omega_0^2}$: the resonant response exceeds the static response by the factor Q .

5.2 Three distinct frequencies

It is worth displaying explicitly the hierarchy that has now emerged — three frequencies that coincide only in the undamped limit and are frequently conflated:

$$\omega_{\text{peak}}^2 = \omega_0^2 - 2\beta^2 < \omega_d^2 = \omega_0^2 - \beta^2 < \omega_0^2 \quad (110)$$

Frequency	Where it appears	Role
ω_0	§2–§4	natural frequency; phase lag crosses $\pi/2$ here
ω_d	§3.3	frequency of <i>free</i> damped oscillation
ω_{peak}	this section	driving frequency of maximum <i>amplitude</i>

(A fourth, the frequency of maximum *velocity* response, sits exactly at ω_0 ; see §5.5.) The differences are $O(\beta^2)$ and thus invisible at weak damping, which is why the distinctions are easy to miss.

5.3 Width of the resonance and the quality factor

Assume weak damping, $\beta \ll \omega_0$, and examine $A(\omega)$ near the peak. For ω near ω_0 , factor the resonant denominator:

$$\omega_0^2 - \omega^2 = (\omega_0 - \omega)(\omega_0 + \omega) \approx 2\omega_0(\omega_0 - \omega), \quad 4\beta^2\omega^2 \approx 4\beta^2\omega_0^2, \quad (111)$$

where in each factor we replaced $\omega \rightarrow \omega_0$ *except* in the difference $\omega_0 - \omega$, which is the small quantity being tracked. Then

$$A^2(\omega) \approx \frac{f_0^2}{4\omega_0^2(\omega_0 - \omega)^2 + 4\beta^2\omega_0^2} = \frac{f_0^2}{4\omega_0^2} \cdot \frac{1}{(\omega - \omega_0)^2 + \beta^2}. \quad (112)$$

As a function of ω , the squared amplitude (equivalently the stored energy) is a *Lorentzian* centered at ω_0 . Its maximum is $\frac{f_0^2}{4\omega_0^2\beta^2}$; it falls to half that value when

$$(\omega - \omega_0)^2 = \beta^2 \quad \Longleftrightarrow \quad \omega = \omega_0 \pm \beta, \quad (113)$$

so the *full width at half maximum* of the energy response is

$$\boxed{\Delta\omega_{\text{FWHM}} = 2\beta \quad \Longrightarrow \quad \frac{\omega_0}{\Delta\omega_{\text{FWHM}}} = \frac{\omega_0}{2\beta} = Q} \quad (114)$$

The same Q defined in §3 from the free decay rate reappears as the dimensionless sharpness of the driven resonance:

$$Q = 2\pi \frac{\text{energy stored}}{\text{energy lost per cycle}} = \frac{\text{center frequency}}{\text{linewidth}}. \quad (115)$$

(The first equality, for weak damping: $E \propto e^{-2\beta t}$ loses $\Delta E \approx 2\beta T E$ per period $T \approx 2\pi/\omega_0$, so $2\pi E/\Delta E = \omega_0/(2\beta) = Q$.) Slow decay in time $\Leftrightarrow$ narrow peak in frequency — a time–frequency tradeoff.

5.4 Phase behavior near resonance

From (89), at $\omega = \omega_0$ the real part of $\hat{D}$ vanishes, so $\delta = \pi/2$ *exactly*, for every β . For the slope, expand the cotangent near ω_0 ($\omega = \omega_0 + \varepsilon$):

$$\cot \delta = \frac{\omega_0^2 - \omega^2}{2\beta\omega} \approx \frac{-2\omega_0\varepsilon}{2\beta\omega_0} = -\frac{\varepsilon}{\beta}. \quad (116)$$

Since $\cot \delta = -\tan(\delta - \frac{\pi}{2})$, and $\tan s \approx s$ for small s ,

$$\delta(\omega) \approx \frac{\pi}{2} + \frac{\omega - \omega_0}{\beta}, \quad \left. \frac{d\delta}{d\omega} \right|_{\omega_0} = \frac{1}{\beta}. \quad (117)$$

The phase sweeps through $\pi/2$ with slope $1/\beta$: the weaker the damping, the more abrupt the crossing (Fig. 5), degenerating into the step function of the undamped Case A.

5.5 Aside: velocity resonance and power absorption

Not the main thread, but worth recording because it explains why “resonance at ω_0 ” is also correct — for a different response variable. The steady-state *velocity* amplitude is $V(\omega) = \omega A(\omega)$. Dividing inside the square root by ω^2 :

$$V(\omega) = \frac{f_0 \omega}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}} = \frac{f_0}{\sqrt{(\frac{\omega_0^2}{\omega} - \omega)^2 + 4\beta^2}}. \quad (118)$$

The only ω -dependence left is the term $(\frac{\omega_0^2}{\omega} - \omega)^2 \geq 0$, which vanishes iff $\omega = \omega_0$: velocity resonance sits *exactly* at ω_0 for any damping. The time-averaged power delivered by the drive,

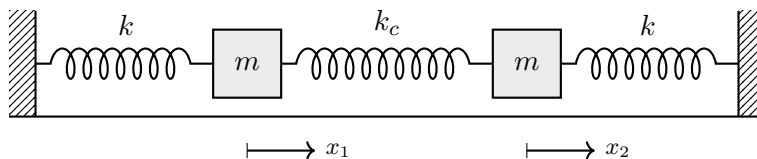
$$\langle P \rangle = \langle F(t) \dot{x}(t) \rangle = \langle F_0 \cos \omega t \cdot (-\omega A \sin(\omega t - \delta)) \rangle = \frac{1}{2} F_0 \omega A \sin \delta, \quad (119)$$

(using $\sin(\omega t - \delta) = \sin \omega t \cos \delta - \cos \omega t \sin \delta$, $\langle \cos \omega t \sin \omega t \rangle = 0$, $\langle \cos^2 \omega t \rangle = \frac{1}{2}$) is likewise maximized at ω_0 , where $\sin \delta = 1$ and V peaks simultaneously. In steady state this input power exactly balances the dissipation $\langle b\dot{x}^2 \rangle$.

6 Coupled Harmonic Oscillators

6.1 Physical setup and equations of motion

Two identical blocks of mass m slide on a frictionless surface between two walls. Each block is attached to its wall by a spring of stiffness k , and the two blocks are connected to each other by a *coupling spring* of stiffness k_c . Let x_1, x_2 be the displacements of blocks 1, 2 from their respective equilibrium positions (both measured positive to the right).



Free-body analysis (the coupling spring is where sign errors happen, so we do it slowly). The *extension* of the coupling spring beyond its relaxed length is

$$\Delta_c = x_2 - x_1 \quad (120)$$

(block 2 moving right stretches it; block 1 moving right compresses it). A stretched spring pulls its two ends toward each other:

- on block 1 it pulls to the *right*: force $+k_c(x_2 - x_1)$;
- on block 2 it pulls to the *left*: force $-k_c(x_2 - x_1)$.

(Check with $\Delta_c < 0$, compressed: the spring pushes the blocks apart, and both formulas flip sign correctly. Also note the two forces are equal and opposite — Newton's third law is built in.) The wall springs contribute $-kx_1$ and $-kx_2$ as before. Newton's second law for each block:

$$\boxed{\begin{aligned} m\ddot{x}_1 &= -kx_1 + k_c(x_2 - x_1), \\ m\ddot{x}_2 &= -kx_2 - k_c(x_2 - x_1). \end{aligned}} \quad (121)$$

The equations are *coupled*: $\ddot{x}_1$ depends on x_2 and vice versa. Neither equation can be solved alone.

Convention 6.1 (Coupling frequency).

$$\boxed{\omega_c^2 \equiv \frac{k_c}{m}} \quad (\text{and as before } \omega_0^2 = k/m). \quad (122)$$

6.2 Matrix form

Divide (121) by m and collect:

$$\begin{aligned}\ddot{x}_1 &= -(\omega_0^2 + \omega_c^2) x_1 + \omega_c^2 x_2, \\ \ddot{x}_2 &= \omega_c^2 x_1 - (\omega_0^2 + \omega_c^2) x_2.\end{aligned}\tag{123}$$

Introduce the column vector $\vec{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$; then

$$\boxed{\ddot{\vec{x}} = -\mathbf{A} \vec{x}, \quad \mathbf{A} = \begin{pmatrix} \omega_0^2 + \omega_c^2 & -\omega_c^2 \\ -\omega_c^2 & \omega_0^2 + \omega_c^2 \end{pmatrix}}\tag{124}$$

This is formally the SHO equation $\ddot{x} = -\omega_0^2 x$ with the scalar ω_0^2 promoted to the real symmetric matrix $\mathbf{A}$. The strategy is clear: find the directions in which $\mathbf{A}$ acts like a scalar — its eigenvectors — and the system will fall apart into independent SHOs.

6.3 Normal modes as an eigenvalue problem

Try a solution in which *both* coordinates oscillate at one common frequency with a fixed amplitude ratio — the same exponential ansatz as always, now vector-valued:

$$\tilde{\vec{x}}(t) = \vec{v} e^{i\omega t}, \quad \vec{v} \in \mathbb{R}^2 \text{ constant}\tag{125}$$

(complex stand-in as usual; the physical motion is its real part $\vec{v} \cos \omega t$). Substituting into (124):

$$-\omega^2 \vec{v} e^{i\omega t} = -\mathbf{A} \vec{v} e^{i\omega t} \iff \boxed{\mathbf{A} \vec{v} = \omega^2 \vec{v}}\tag{126}$$

— an eigenvalue problem: the allowed frequencies squared are the eigenvalues of $\mathbf{A}$, and the corresponding motion patterns are its eigenvectors. Nontrivial $\vec{v}$ exists iff

$$\det(\mathbf{A} - \omega^2 \mathbb{I}) = 0.\tag{127}$$

Compute, abbreviating $s \equiv \omega_0^2 + \omega_c^2$:

$$\det \begin{pmatrix} s - \omega^2 & -\omega_c^2 \\ -\omega_c^2 & s - \omega^2 \end{pmatrix} = (s - \omega^2)^2 - \omega_c^4 = 0 \iff s - \omega^2 = \pm \omega_c^2.\tag{128}$$

Hence the two *normal frequencies*:

$$\boxed{\omega_1^2 = s - \omega_c^2 = \omega_0^2, \quad \omega_2^2 = s + \omega_c^2 = \omega_0^2 + 2\omega_c^2}\tag{129}$$

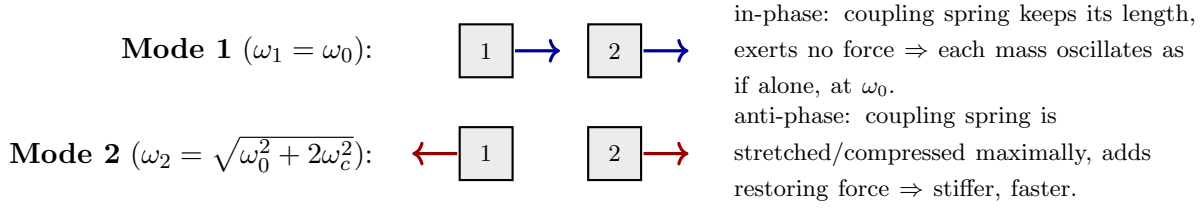
Eigenvectors: for $\omega^2 = \omega_1^2$, the equation $(\mathbf{A} - \omega_1^2 \mathbb{I})\vec{v} = 0$ reads

$$\begin{pmatrix} \omega_c^2 & -\omega_c^2 \\ -\omega_c^2 & \omega_c^2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0 \implies v_1 = v_2 \implies \vec{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix};\tag{130}$$

for $\omega^2 = \omega_2^2$,

$$\begin{pmatrix} -\omega_c^2 & -\omega_c^2 \\ -\omega_c^2 & -\omega_c^2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0 \implies v_1 = -v_2 \implies \vec{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.\tag{131}$$

Physical interpretation.



That mode 1 comes out at exactly ω_0 , untouched by k_c , and that mode 2 sees the coupling spring *twice* ($2\omega_c^2$: each mass moves against a spring whose other end is moving the opposite way, doubling the relative displacement) are both visible in the picture before any algebra.

6.4 Decoupling by a change of coordinates

The eigenvector directions suggest new coordinates. Define the *normal coordinates*

$$\boxed{q_1 \equiv \frac{x_1 + x_2}{2}, \quad q_2 \equiv \frac{x_1 - x_2}{2}} \iff x_1 = q_1 + q_2, \quad x_2 = q_1 - q_2. \quad (132)$$

(The normalization $\frac{1}{2}$ is a convention; it makes the inverse map coefficient-free.) Now *add* and *subtract* the two equations (121) divided by m :

$$\begin{aligned} \text{sum:} \quad \ddot{x}_1 + \ddot{x}_2 &= -\omega_0^2(x_1 + x_2) + \omega_c^2(x_2 - x_1) - \omega_c^2(x_2 - x_1) = -\omega_0^2(x_1 + x_2), \\ \text{difference:} \quad \ddot{x}_1 - \ddot{x}_2 &= -\omega_0^2(x_1 - x_2) + \omega_c^2(x_2 - x_1) + \omega_c^2(x_2 - x_1) = -(\omega_0^2 + 2\omega_c^2)(x_1 - x_2). \end{aligned} \quad (133)$$

Dividing by 2, the equations for q_1, q_2 close on themselves:

$$\boxed{\ddot{q}_1 + \omega_1^2 q_1 = 0, \quad \ddot{q}_2 + \omega_2^2 q_2 = 0} \quad (\omega_1^2 = \omega_0^2, \quad \omega_2^2 = \omega_0^2 + 2\omega_c^2). \quad (134)$$

The coupled pair has become two *independent* simple harmonic oscillators — everything from §2 applies verbatim to each. In matrix language: the change of basis to eigenvector coordinates has diagonalized A ,

$$A = S \begin{pmatrix} \omega_1^2 & 0 \\ 0 & \omega_2^2 \end{pmatrix} S^{-1}, \quad S = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad (135)$$

and in the new coordinates the matrix equation $\ddot{\vec{x}} = -A\vec{x}$ falls apart row by row.

6.5 General solution and initial conditions

Each normal coordinate carries the full SHO solution (16):

$$q_1(t) = A_1 \cos(\omega_1 t - \phi_1), \quad q_2(t) = A_2 \cos(\omega_2 t - \phi_2), \quad (136)$$

so the general motion of the blocks is the superposition

$$\boxed{\begin{aligned} x_1(t) &= A_1 \cos(\omega_1 t - \phi_1) + A_2 \cos(\omega_2 t - \phi_2), \\ x_2(t) &= A_1 \cos(\omega_1 t - \phi_1) - A_2 \cos(\omega_2 t - \phi_2), \end{aligned}} \quad (137)$$

with four constants (A_1, ϕ_1, A_2, ϕ_2) matching the four initial data $x_1(0), x_2(0), \dot{x}_1(0), \dot{x}_2(0)$. Generic motion is *not* periodic (unless ω_2/ω_1 is rational): only the pure modes oscillate at a single frequency.

6.6 Worked example: energy exchange (beats again)

Displace block 1 by a and release everything from rest:

$$x_1(0) = a, \quad x_2(0) = 0, \quad \dot{x}_1(0) = \dot{x}_2(0) = 0. \quad (138)$$

In normal coordinates: $q_1(0) = \frac{a}{2}$, $q_2(0) = \frac{a}{2}$, $\dot{q}_1(0) = \dot{q}_2(0) = 0$. Each mode starts at rest, so ((11) with $v_0 = 0$) $q_i(t) = q_i(0) \cos \omega_i t$:

$$q_1 = \frac{a}{2} \cos \omega_1 t, \quad q_2 = \frac{a}{2} \cos \omega_2 t. \quad (139)$$

Transforming back:

$$x_1(t) = \frac{a}{2} (\cos \omega_1 t + \cos \omega_2 t), \quad x_2(t) = \frac{a}{2} (\cos \omega_1 t - \cos \omega_2 t). \quad (140)$$

Apply the product identities (sum and difference forms; the difference form was derived in §4.1, the sum form follows the same way from adding the two addition formulas):

$$\boxed{x_1(t) = a \cos\left(\frac{\omega_2 - \omega_1}{2} t\right) \cos\left(\frac{\omega_1 + \omega_2}{2} t\right), \quad x_2(t) = a \sin\left(\frac{\omega_2 - \omega_1}{2} t\right) \sin\left(\frac{\omega_1 + \omega_2}{2} t\right)} \quad (141)$$

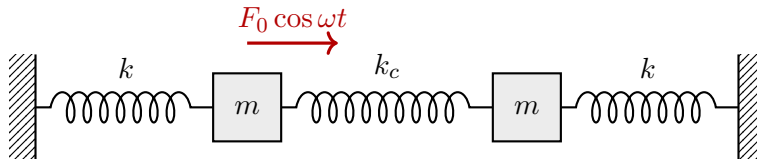
For weak coupling ($\omega_c \ll \omega_0$, so $\omega_2 - \omega_1 \approx \omega_c^2/\omega_0$ is small), both blocks oscillate fast at $\approx \omega_0$ inside *slow, complementary* envelopes: $|\cos|$ for block 1 and $|\sin|$ for block 2 (Fig. 6). When block 1's envelope is at a node, block 2's is at an antinode — the energy sloshes back and forth through the coupling spring with period $2\pi/(\omega_2 - \omega_1)$, complete transfer occurring precisely because the two modes were excited with equal amplitude.

Remark 6.2 (Why this always works). Nothing here depended on luck. $\mathbf{A}$ is real and symmetric (this traces back to Newton's third law acting through the coupling spring), and every real symmetric matrix is diagonalizable by an orthogonal change of basis with real eigenvalues [stated; the spectral theorem of linear algebra]; positive-definiteness of $\mathbf{A}$ (all spring constants positive) makes the eigenvalues positive, so each is legitimately an ω_i^2 . The recipe — write $\ddot{\vec{x}} = -\mathbf{A}\vec{x}$, find eigenpairs, decouple — therefore extends to N masses, unequal parameters, and beyond. For unequal masses one writes $\mathbf{M}\ddot{\vec{x}} = -\mathbf{K}\vec{x}$ and diagonalizes $\mathbf{M}^{-1}\mathbf{K}$ (symmetric with respect to the mass-weighted inner product) [stated; same theorem, adapted inner product]; we will not need this generality here.

7 The Forced Coupled Oscillator

7.1 Setup

Drive *block 1 only* with the force $F_0 \cos \omega t$; block 2 is driven purely through the coupling. We treat the undamped system in the main text (formulas stay exact and transparent); the figures overlay a small damping to show what a real measurement would produce.



Adding $f_0 \cos \omega t$ (with $f_0 = F_0/m$) to the first equation of motion:

$$\boxed{\ddot{\vec{x}} + \mathbf{A} \vec{x} = f_0 \cos \omega t \vec{e}_1, \quad \vec{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}} \quad (142)$$

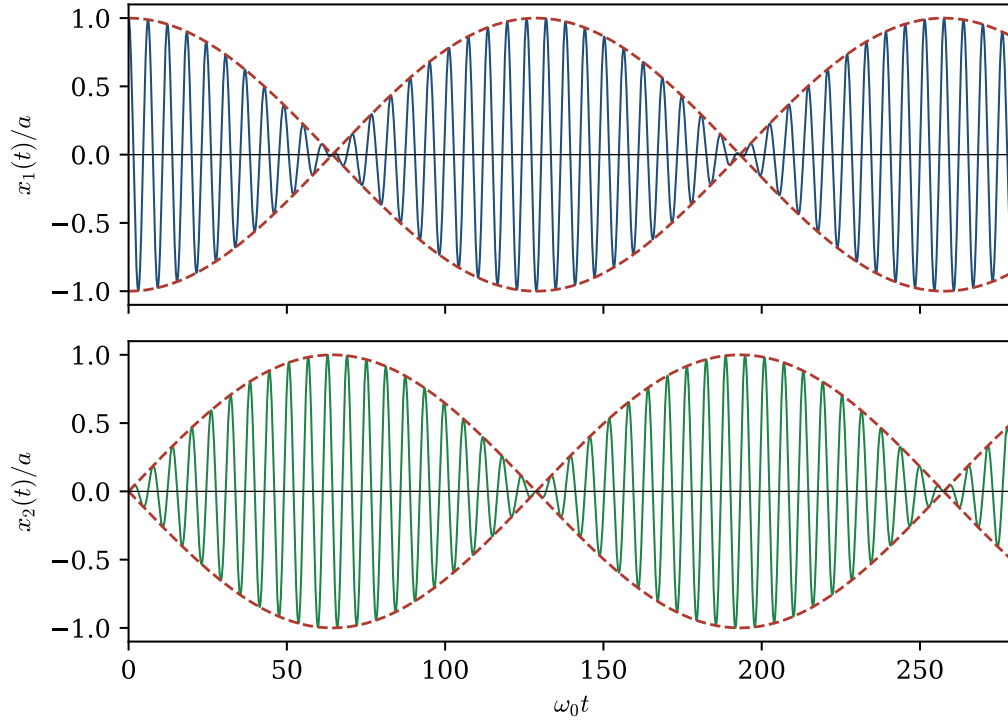


Figure 6: Weakly coupled oscillators ($\omega_c^2 = 0.05 \omega_0^2$), block 1 displaced and released from rest. The envelopes (dashed) are complementary: energy migrates fully from block 1 to block 2 and back.

7.2 Reduction to two independent forced oscillators

Pass to the normal coordinates of §6.4. Add and subtract the two component equations of (142) (only the first has a drive) and divide by 2:

$$\begin{aligned} \frac{1}{2}(\text{eq}_1 + \text{eq}_2) : \quad & \ddot{q}_1 + \omega_1^2 q_1 = \frac{f_0}{2} \cos \omega t, \\ \frac{1}{2}(\text{eq}_1 - \text{eq}_2) : \quad & \ddot{q}_2 + \omega_2^2 q_2 = \frac{f_0}{2} \cos \omega t. \end{aligned} \quad (143)$$

Interpretation: a force applied to block 1 alone decomposes as $\vec{e}_1 = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ — it pushes on *both* modes, each with half strength. Each line of (143) is exactly Case A of §4.1, already solved: the steady-state (particular) responses are

$$q_{1,p}(t) = \frac{f_0/2}{\omega_1^2 - \omega^2} \cos \omega t, \quad q_{2,p}(t) = \frac{f_0/2}{\omega_2^2 - \omega^2} \cos \omega t. \quad (144)$$

Every piece of machinery built in §4 is being reused; the only new step was the change of basis.

7.3 Response of the physical coordinates; two resonances

Transform back ($x_1 = q_1 + q_2$, $x_2 = q_1 - q_2$):

$$x_{1,p}(t) = X_1(\omega) \cos \omega t, \quad x_{2,p}(t) = X_2(\omega) \cos \omega t, \quad (145)$$

with the response functions

$$\boxed{X_1(\omega) = \frac{f_0}{2} \left[\frac{1}{\omega_1^2 - \omega^2} + \frac{1}{\omega_2^2 - \omega^2} \right], \quad X_2(\omega) = \frac{f_0}{2} \left[\frac{1}{\omega_1^2 - \omega^2} - \frac{1}{\omega_2^2 - \omega^2} \right]} \quad (146)$$

Both responses diverge at $\omega = \omega_1$ and at $\omega = \omega_2$: driving at either normal frequency resonantly excites the corresponding mode, and both blocks participate in each mode. *The resonances of a coupled system sit at its normal frequencies, not at the frequencies of its parts.* With small damping the divergences become the two finite peaks in Fig. 7.

7.4 Antiresonance

Between the two resonances, something less familiar happens to the *driven* block. Combine the two fractions in X_1 :

$$X_1(\omega) = \frac{f_0}{2} \cdot \frac{(\omega_2^2 - \omega^2) + (\omega_1^2 - \omega^2)}{(\omega_1^2 - \omega^2)(\omega_2^2 - \omega^2)} = \frac{f_0}{2} \cdot \frac{\omega_1^2 + \omega_2^2 - 2\omega^2}{(\omega_1^2 - \omega^2)(\omega_2^2 - \omega^2)}. \quad (147)$$

The numerator vanishes at

$$\omega_a^2 = \frac{\omega_1^2 + \omega_2^2}{2} = \frac{\omega_0^2 + (\omega_0^2 + 2\omega_c^2)}{2} \implies \boxed{\omega_a = \sqrt{\omega_0^2 + \omega_c^2}} \quad (148)$$

which lies strictly between ω_1 and ω_2 . At this *antiresonance* frequency, the block being pushed *does not move at all* (in steady state), while the undriven block does. Indeed, evaluating X_2 at ω_a : since $\omega_1^2 - \omega_a^2 = -\omega_c^2$ and $\omega_2^2 - \omega_a^2 = +\omega_c^2$,

$$X_2(\omega_a) = \frac{f_0}{2} \left[\frac{1}{-\omega_c^2} - \frac{1}{\omega_c^2} \right] = -\frac{f_0}{\omega_c^2} \neq 0. \quad (149)$$

How can the driven block stand still? Check force balance on block 1 directly. If $x_1 \equiv 0$ and $x_2 = -\frac{f_0}{\omega_c^2} \cos \omega t$, the coupling spring exerts on block 1 the force

$$k_c(x_2 - x_1) = k_c x_2 = m\omega_c^2 \cdot \left(-\frac{f_0}{\omega_c^2}\right) \cos \omega t = -mf_0 \cos \omega t = -F_0 \cos \omega t, \quad (150)$$

which cancels the applied drive *exactly*, at every instant. Block 2 has spontaneously arranged itself in antiphase with just the right amplitude to shield block 1 from its own drive. (Consistency: with $x_1 \equiv 0$, block 2 is a free oscillator bound by springs k and k_c in parallel, natural frequency $\sqrt{(k + k_c)/m} = \sqrt{\omega_0^2 + \omega_c^2} = \omega_a$ — the antiresonance of block 1 is the natural frequency of block 2 *with block 1 clamped*.)

This is the operating principle of the *dynamic vibration absorber* (tuned mass damper): to protect a structure from vibration at a known frequency, attach an auxiliary mass–spring system tuned to that frequency, and it absorbs the motion. The skyscraper pendulum is this calculation at scale.

Remark 7.1 (On the undamped idealization). In the undamped system the homogeneous solutions (free oscillations of the two modes) never decay, so strictly there is no unique long-time steady state: the particular solution above is *the* steady state only in the presence of some damping, however small, which kills the transients exactly as in §4.2.2 — mode by mode. This is the sense in which the undamped formulas (146) should be read: as the $\beta \rightarrow 0^+$ limit of the damped response, which is what the dashed curves of Fig. 7 converge to.

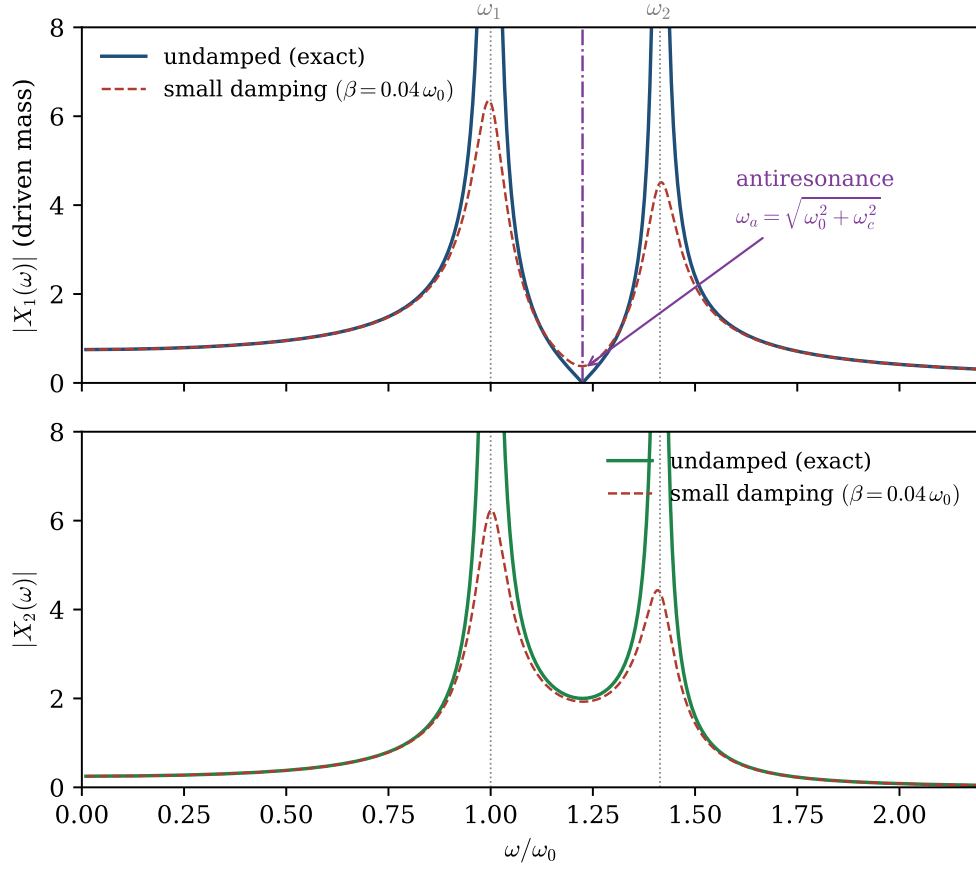


Figure 7: Steady-state amplitudes versus driving frequency for the forced coupled pair ($\omega_c^2 = 0.5\omega_0^2$, so $\omega_1 = \omega_0$, $\omega_2 = \sqrt{2}\omega_0$, $\omega_a = \sqrt{1.5}\omega_0$). Solid: exact undamped response (146) (clipped near the divergences). Dashed: the same system with small damping on both blocks, computed numerically — peaks become finite, the antiresonance zero of $|X_1|$ becomes a sharp dip.

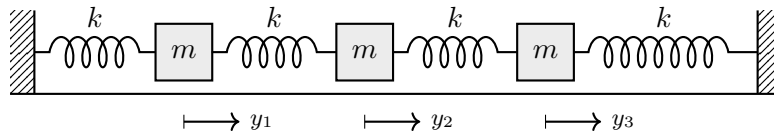
8 The N -Mass Chain

The two-mass system generalizes to a chain of N identical masses. From this section on we make two adjustments. First, *all* springs are taken identical with stiffness k (the special case $k_c = k$ of §6); $\omega_0^2 = k/m$ as always. Second, we rename the displacements $x_j \rightarrow y_j$, reserving the letter x for *position along the chain* — we will need it in the continuum limit of §9.

We build the general case by writing out $N = 3, 4, 5$ explicitly and letting the pattern emerge.

8.1 Three, four, five masses

$N = 3$. Three masses, four springs, walls at both ends:



Free-body analysis is the same computation as in §6, mass by mass. Each mass feels its two neighboring springs; the spring between sites j and $j + 1$ has extension $y_{j+1} - y_j$ and pulls mass j with force $+k(y_{j+1} - y_j)$ and mass $j + 1$ with force $-k(y_{j+1} - y_j)$. The end masses feel a wall spring, e.g. $-ky_1$ on mass 1. Dividing by m :

$$\begin{aligned} \ddot{y}_1 &= \omega_0^2 (-2y_1 + y_2), \\ \ddot{y}_2 &= \omega_0^2 (y_1 - 2y_2 + y_3), \\ \ddot{y}_3 &= \omega_0^2 (y_2 - 2y_3), \end{aligned} \quad \mathbf{A}_3 = \omega_0^2 \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}. \quad (151)$$

$N = 4$. Identical reasoning, one more interior mass:

$$\begin{aligned} \ddot{y}_1 &= \omega_0^2 (-2y_1 + y_2), \\ \ddot{y}_2 &= \omega_0^2 (y_1 - 2y_2 + y_3), \\ \ddot{y}_3 &= \omega_0^2 (y_2 - 2y_3 + y_4), \\ \ddot{y}_4 &= \omega_0^2 (y_3 - 2y_4), \end{aligned} \quad \mathbf{A}_4 = \omega_0^2 \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}. \quad (152)$$

$N = 5$.

$$\begin{aligned} \ddot{y}_1 &= \omega_0^2 (-2y_1 + y_2), \\ \ddot{y}_2 &= \omega_0^2 (y_1 - 2y_2 + y_3), \\ \ddot{y}_3 &= \omega_0^2 (y_2 - 2y_3 + y_4), \\ \ddot{y}_4 &= \omega_0^2 (y_3 - 2y_4 + y_5), \\ \ddot{y}_5 &= \omega_0^2 (y_4 - 2y_5), \end{aligned} \quad \mathbf{A}_5 = \omega_0^2 \begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{pmatrix}. \quad (153)$$

The pattern. Three observations, each visible above:

1. Every *interior* mass obeys the same equation, $\ddot{y}_j = \omega_0^2 (y_{j-1} - 2y_j + y_{j+1})$: the chain is translation-invariant in its bulk.
2. The *end* equations look truncated — but they are the interior equation in disguise. Introduce *phantom sites* $j = 0$ and $j = N + 1$ at the walls and declare

$$\boxed{y_0(t) \equiv 0, \quad y_{N+1}(t) \equiv 0} \quad (\text{fixed-wall boundary conditions}). \quad (154)$$

Then $\ddot{y}_1 = \omega_0^2 (y_0 - 2y_1 + y_2)$ reproduces the first row exactly, and likewise at the far end. One rule, all rows.

3. The matrix is *tridiagonal*: 2's on the diagonal, -1 's on the two adjacent diagonals, zeros elsewhere — each mass talks only to its neighbors.

8.2 The general chain

The induction is now immediate: for any N , the equations of motion are

$$\boxed{\ddot{y}_j = \omega_0^2 (y_{j-1} - 2y_j + y_{j+1}), \quad j = 1, \dots, N, \quad y_0 = y_{N+1} = 0} \quad (155)$$

or in matrix form $\ddot{\vec{y}} = -\mathbf{A}_N \vec{y}$ with

$$\mathbf{A}_N = \omega_0^2 \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix}_{N \times N} \quad (\text{blank entries are } 0). \quad (156)$$

$\mathbf{A}_N$ is real symmetric, so the program of §6 applies: N normal modes with real frequencies, mutually decoupled. For $N = 2$ the determinant was a quadratic; for general N we need a better idea than expanding an $N \times N$ determinant.

8.3 Normal modes: an exponential ansatz in space

Substituting $\tilde{\vec{y}}(t) = \vec{v} e^{i\omega t}$ as in §6, the eigenvalue problem $\mathbf{A}_N \vec{v} = \omega^2 \vec{v}$ reads, component by component,

$$\omega_0^2 (2v_j - v_{j-1} - v_{j+1}) = \omega^2 v_j, \quad j = 1, \dots, N, \quad v_0 = v_{N+1} = 0. \quad (157)$$

Look at what this is: a *linear recursion in j with constant coefficients* — the same structure our differential equations had in t , with the site index playing the role of time. In §2.4 the winning move for such equations was the exponential ansatz; try its discrete counterpart, a geometric sequence in the site index:

$$v_j = e^{ij\theta}, \quad \theta \text{ constant}. \quad (158)$$

Substitute into the interior equation (157) and divide by $e^{ij\theta} \neq 0$:

$$\omega_0^2 (2 - e^{-i\theta} - e^{+i\theta}) = \omega^2 \implies \omega^2 = 2\omega_0^2 (1 - \cos \theta), \quad (159)$$

using $e^{i\theta} + e^{-i\theta} = 2 \cos \theta$ (23). With the half-angle identity $1 - \cos \theta = 2 \sin^2(\theta/2)$:

$$\boxed{\omega^2 = 4\omega_0^2 \sin^2 \frac{\theta}{2}} \quad (160)$$

This is the *dispersion relation* of the chain: it converts a spatial oscillation rate θ (radians of phase per site) into a temporal frequency ω . Note what has and has not been fixed: *every* θ solves the interior recursion — the analog of “every root of the characteristic equation gives a solution.” It is the *boundary conditions* that select the allowed values, exactly as initial conditions selected constants before.

Imposing the boundaries. Since ω^2 in (160) depends only on $\cos \theta$, the two sequences $e^{\pm ij\theta}$ carry the *same* frequency, and any combination of them still solves the interior recursion. The combination

$$v_j = \frac{e^{ij\theta} - e^{-ij\theta}}{2i} = \sin(j\theta) \quad (161)$$

satisfies the left boundary automatically: $v_0 = \sin 0 = 0$. The right boundary then quantizes θ :

$$v_{N+1} = \sin((N+1)\theta) = 0 \iff (N+1)\theta = n\pi \iff \boxed{\theta_n = \frac{n\pi}{N+1}} \quad (162)$$

with n an integer. Which integers give genuinely distinct modes? $n = 0$ and $n = N + 1$ make $v_j = \sin(j\theta_n)$ vanish identically (all sites at a node); $n' = 2(N + 1) - n$ gives $\sin(j\theta_{n'}) = -\sin(j\theta_n)$, the same mode up to overall sign; larger n repeats periodically. The distinct modes are exactly

$$n = 1, 2, \dots, N, \quad (163)$$

— N of them, matching the dimension of the problem, so we have found *all* the normal modes without ever expanding a determinant:

$$\boxed{v_j^{(n)} = \sin\left(\frac{jn\pi}{N+1}\right), \quad \omega_n = 2\omega_0 \sin\left(\frac{n\pi}{2(N+1)}\right), \quad n = 1, \dots, N} \quad (164)$$

Consistency check against $N = 2$. Formula (164) with $N = 2$ (so $\theta_n = n\pi/3$):

$$\omega_1^2 = 4\omega_0^2 \sin^2 \frac{\pi}{6} = 4\omega_0^2 \cdot \frac{1}{4} = \omega_0^2, \quad \vec{v}^{(1)} = \left(\sin \frac{\pi}{3}, \sin \frac{2\pi}{3}\right) = \frac{\sqrt{3}}{2} (1, 1), \quad (165)$$

$$\omega_2^2 = 4\omega_0^2 \sin^2 \frac{\pi}{3} = 4\omega_0^2 \cdot \frac{3}{4} = 3\omega_0^2, \quad \vec{v}^{(2)} = \left(\sin \frac{2\pi}{3}, \sin \frac{4\pi}{3}\right) = \frac{\sqrt{3}}{2} (1, -1). \quad (166)$$

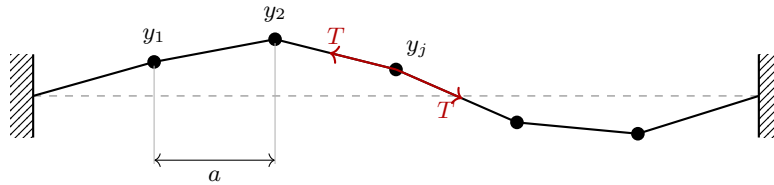
Compare §6 with $k_c = k$: there $\omega_1^2 = \omega_0^2$ and $\omega_2^2 = \omega_0^2 + 2\omega_c^2 = 3\omega_0^2$, with modes $(1, 1)$ and $(1, -1)$. Exact agreement.

Remark 8.1 (Reading the mode shapes). Low n : long, smooth waves; all the low-frequency physics. High n : rapidly alternating displacements, approaching the zigzag pattern at $\theta = \pi$ where each mass moves opposite to its neighbors and every spring is maximally worked — hence the maximum frequency $2\omega_0$. The chain has a *fastest* oscillation; there is no pattern it can support above $2\omega_0$. A continuous string will have no such ceiling, and comparing the two limits is instructive: the ceiling is the memory of the graininess.

9 The Continuum Limit: From Chain to Rope

9.1 The rope as a bead chain

The chain of §8 was longitudinal (masses displace along the line of springs). The classic continuum system — a stretched rope or string — is *transverse*, and it turns out to be the same mathematics. Consider N beads of mass m , spaced a apart on a massless string stretched to tension T between fixed walls; let y_j now denote the small *transverse* displacement of bead j .



Transverse force on bead j . The string segment from bead j to bead $j + 1$ makes a small angle φ_{j+} with the horizontal, with $\tan \varphi_{j+} = (y_{j+1} - y_j)/a$. It pulls bead j along itself with magnitude T ; the transverse component is

$$T \sin \varphi_{j+} \approx T \tan \varphi_{j+} = \frac{T}{a} (y_{j+1} - y_j) \quad (167)$$

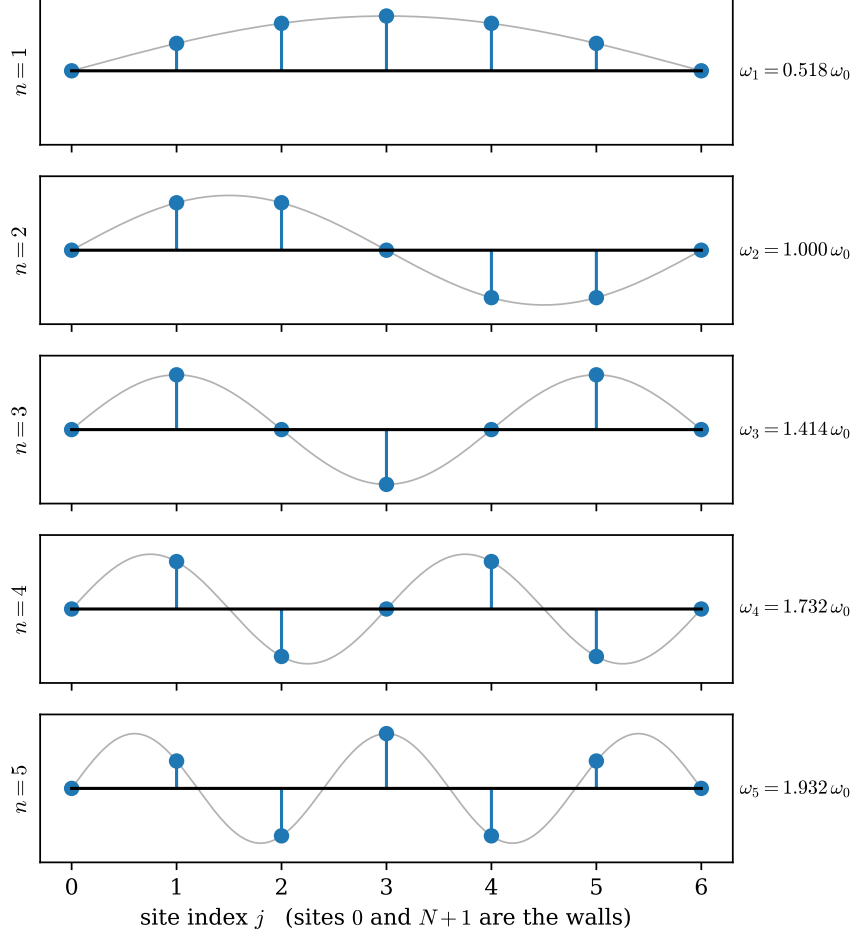


Figure 8: The five normal modes of the $N = 5$ chain. Dots: the mode vector components $v_j^{(n)} = \sin(jn\pi/6)$ at the sites (the phantom wall sites $j = 0, 6$ included). Gray curve: the interpolating function $\sin(n\pi x/6)$ — the masses sample a smooth standing wave, a fact that becomes the whole story in §9.

(small angles: $\sin \varphi \approx \tan \varphi \approx \varphi$; to this order the tension itself is unchanged). The left segment likewise contributes $\frac{T}{a}(y_{j-1} - y_j)$. Newton's second law transversely:

$$m\ddot{y}_j = \frac{T}{a}(y_{j-1} - 2y_j + y_{j+1}). \quad (168)$$

This is exactly (155) with the substitution

$$\boxed{k \longrightarrow \frac{T}{a}, \quad \omega_0^2 = \frac{T}{ma}} \quad (169)$$

Everything from §8 — modes, dispersion, the $2\omega_0$ ceiling — transfers verbatim. From here on we work with the beads.

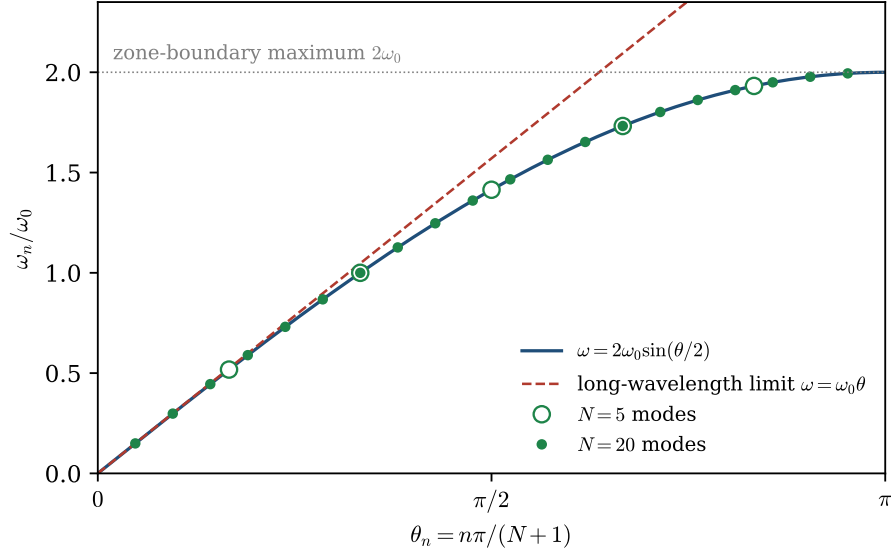


Figure 9: The dispersion relation (160). Allowed modes sit at $\theta_n = n\pi/(N+1)$: shown for $N = 5$ (open circles) and $N = 20$ (filled dots) — as N grows the modes fill in the curve. Dashed: the long-wavelength tangent $\omega = \omega_0\theta$, the seed of the continuum limit. The curve saturates at $2\omega_0$ at $\theta = \pi$, where adjacent masses move in exact opposition — the stiffest, fastest pattern the chain supports.

9.2 Taking the limit

Refine the chain: more beads, closer together, keeping the *physical* rope fixed. Precisely, let

$$a \rightarrow 0, \quad N \rightarrow \infty, \quad \text{holding fixed } L = (N+1)a \text{ (length)}, \quad \mu = \frac{m}{a} \text{ (mass per length)}, \quad T. \quad (170)$$

Each bead's mass $m = \mu a \rightarrow 0$; the discrete displacements merge into a field: postulate a smooth function $y(x, t)$ interpolating them,

$$y_j(t) = y(x_j, t), \quad x_j = j a. \quad (171)$$

The neighbors sit at $x_j \pm a$; Taylor-expand in the small spacing:

$$y_{j\pm 1} = y(x_j \pm a, t) = y \pm a \frac{\partial y}{\partial x} + \frac{a^2}{2} \frac{\partial^2 y}{\partial x^2} \pm \frac{a^3}{6} \frac{\partial^3 y}{\partial x^3} + O(a^4), \quad (172)$$

all derivatives evaluated at (x_j, t) . Add the two expansions: the odd powers of a cancel in pairs,

$$y_{j+1} + y_{j-1} - 2y_j = a^2 \frac{\partial^2 y}{\partial x^2} + O(a^4). \quad (173)$$

(The combination $y_{j-1} - 2y_j + y_{j+1}$ that has followed us since the $N = 3$ matrix is a *second difference*: it was the discrete second derivative all along, and the tridiagonal matrix was the second-derivative operator in disguise.) Substitute into (168) and divide by $m = \mu a$:

$$\frac{\partial^2 y}{\partial t^2} = \frac{T}{\mu a^2} \left(a^2 \frac{\partial^2 y}{\partial x^2} + O(a^4) \right) = \frac{T}{\mu} \frac{\partial^2 y}{\partial x^2} + O(a^2). \quad (174)$$

As $a \rightarrow 0$ the correction vanishes and, defining the constant

$$\boxed{c^2 \equiv \frac{T}{\mu}} \quad [c] = \sqrt{\frac{[\text{force}]}{[\text{mass}]/[\text{length}]} = \frac{[\text{length}]}{[\text{time}]}, \quad (175)$$

we obtain the *wave equation*:

$$\boxed{\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}} \quad y(0, t) = y(L, t) = 0. \quad (176)$$

The boundary conditions are the phantom sites, carried into the limit: $y_0 = 0$ at $x_0 = 0$ and $y_{N+1} = 0$ at $x_{N+1} = L$. (For the longitudinal spring chain the identical computation gives $c^2 = ka^2/m$; only the expression for c in terms of microscopic parameters changes, not the equation.)

9.3 What happens to the modes

Track each object of §8.3 through the limit. Define the *wavenumber* carried by mode n ,

$$k_n \equiv \frac{\theta_n}{a} = \frac{n\pi}{(N+1)a} = \frac{n\pi}{L} \quad [\text{radians of phase per unit length}], \quad (177)$$

which stays *finite* in the limit — $\theta_n \rightarrow 0$ and $a \rightarrow 0$ at matched rates for fixed n .

Mode shapes.

$$v_j^{(n)} = \sin\left(\frac{jn\pi}{N+1}\right) = \sin\left(\frac{n\pi(ja)}{L}\right) = \sin(k_n x_j) \longrightarrow \sin\left(\frac{n\pi x}{L}\right) : \quad (178)$$

the sampled sinusoids of Fig. 8 become the standing waves of a string — n counts the antinodes.

Frequencies. Expand the dispersion relation (160) for small argument, $\sin \varepsilon = \varepsilon + O(\varepsilon^3)$:

$$\omega_n = 2\omega_0 \sin\left(\frac{k_n a}{2}\right) = 2\sqrt{\frac{T}{\mu a^2}} \left(\frac{k_n a}{2} + O(a^3)\right) = \sqrt{\frac{T}{\mu}} k_n + O(a^2) \longrightarrow c k_n, \quad (179)$$

where $\omega_0^2 = T/(ma) = T/(\mu a^2)$ was used. The dispersion relation *linearizes*:

$$\boxed{\omega_n = c k_n = \frac{n\pi c}{L}, \quad n = 1, 2, 3, \dots} \quad (180)$$

This is the harmonic series — overtones at integer multiples of the fundamental $\pi c/L$, the arithmetic that underlies music. On Fig. 9 this is the dashed tangent line at the origin: the continuum keeps only the long-wavelength foot of the chain's dispersion curve, where the graininess is invisible. The frequency ceiling $2\omega_0 = 2c/a$ escapes to infinity as $a \rightarrow 0$: the ideal string has arbitrarily fast modes, and any real string reveals its microscopic structure precisely where the ideal model puts none.

9.4 General solution and the Fourier expansion

Superposition of modes. Each standing wave, set oscillating at its own frequency, solves the wave equation: for any single n ,

$$y_n(x, t) = \sin(k_n x) [A_n \cos \omega_n t + B_n \sin \omega_n t], \quad \omega_n = c k_n, \quad (181)$$

satisfies (176) — verify in one line: ∂_t^2 produces a factor $-\omega_n^2$, $c^2 \partial_x^2$ produces $-c^2 k_n^2$, and these are equal by the dispersion relation; the boundary conditions hold term by term since $\sin(k_n \cdot 0) = \sin(k_n L) = 0$. The wave equation is linear and homogeneous, so superposition applies exactly as it has since §2.2, and the general solution is

$$y(x, t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{L}\right) [A_n \cos \omega_n t + B_n \sin \omega_n t] \quad (182)$$

— the continuum version of “general motion = sum over normal modes,” now with infinitely many modes. In the chain this expansion was guaranteed by linear algebra: $\mathbf{A}_N$ is real symmetric, its eigenvectors span $\mathbb{R}^N$, so *any* initial configuration is a combination of modes. The new question in the continuum is whether that survives $N \rightarrow \infty$.

Initial data. Evaluate (182) and its time derivative at $t = 0$. If the string starts with shape $f(x)$ and velocity profile $g(x)$,

$$f(x) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right), \quad g(x) = \sum_{n=1}^{\infty} B_n \omega_n \sin\left(\frac{n\pi x}{L}\right). \quad (183)$$

Fitting the motion to initial data therefore requires expanding *arbitrary* functions in the mode shapes. That this is possible — every sufficiently well-behaved f with $f(0) = f(L) = 0$ admits such an expansion — is *Fourier’s theorem* [stated; the completeness proof belongs to analysis, not to this document], and the right-hand side of (183) is the *Fourier sine series* of f . What we *can* prove here is how to compute the coefficients, and that the answer is unique.

Orthogonality. The key computation. For integers $n, m \geq 1$, use the product-to-sum identity $\sin P \sin Q = \frac{1}{2}[\cos(P - Q) - \cos(P + Q)]$:

$$\int_0^L \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx = \frac{1}{2} \int_0^L \left[\cos\left(\frac{(n-m)\pi x}{L}\right) - \cos\left(\frac{(n+m)\pi x}{L}\right) \right] dx. \quad (184)$$

For any nonzero integer p , $\int_0^L \cos(p\pi x/L) dx = \left[\frac{L}{p\pi} \sin(p\pi x/L) \right]_0^L = 0$ since $\sin(p\pi) = 0$. If $n \neq m$ both cosine integrals vanish. If $n = m$ the first integrand is $\cos 0 = 1$, giving L , while the second still vanishes. Hence

$$\int_0^L \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx = \frac{L}{2} \delta_{nm} \quad (185)$$

The mode shapes are *orthogonal* — the continuum analog of the orthogonality of eigenvectors of the symmetric matrix $\mathbf{A}_N$, with the integral playing the role of the dot product. (Indeed the chain modes obey the discrete identity $\sum_{j=1}^N \sin(j\theta_n) \sin(j\theta_m) = \frac{N+1}{2} \delta_{nm}$ [stated; it follows from the same product-to-sum identity applied to finite geometric sums, and nothing below depends on it]. Multiplying by the spacing a and letting $a \rightarrow 0$ turns the sum into the integral and $\frac{(N+1)a}{2}$ into $\frac{L}{2}$ — the two statements are one statement.)

Extracting the coefficients. Multiply the first equation of (183) by $\sin(m\pi x/L)$ and integrate over $[0, L]$; orthogonality kills every term of the sum except $n = m$:

$$\int_0^L f(x) \sin\left(\frac{m\pi x}{L}\right) dx = \sum_{n=1}^{\infty} A_n \cdot \frac{L}{2} \delta_{nm} = \frac{L}{2} A_m, \quad (186)$$

so

$$\boxed{A_m = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{m\pi x}{L}\right) dx, \quad B_m = \frac{2}{L\omega_m} \int_0^L g(x) \sin\left(\frac{m\pi x}{L}\right) dx} \quad (187)$$

(the B_m formula obtained the same way from the second equation). Each mode's amplitude is read off by projecting the initial data onto that mode — exactly the operation “decompose the initial condition into normal coordinates” of §6.6, promoted to infinitely many coordinates. Uniqueness is automatic: the coefficients are *determined* by these integrals. Physically, the A_n of a plucked string decide how much of each overtone sounds — the Fourier coefficients of the pluck shape *are* the timbre.

9.5 Traveling waves: d'Alembert's form

The mode expansion answered the fixed-end problem. The wave equation has a second canonical face, best seen with the boundaries removed (the infinite line). Factor the operator,

$$\frac{\partial^2}{\partial t^2} - c^2 \frac{\partial^2}{\partial x^2} = \left(\frac{\partial}{\partial t} - c \frac{\partial}{\partial x}\right) \left(\frac{\partial}{\partial t} + c \frac{\partial}{\partial x}\right), \quad (188)$$

and pass to the characteristic variables $\xi = x - ct$, $\eta = x + ct$. By the chain rule, $\partial_\xi = \frac{1}{2}(\partial_x - c^{-1}\partial_t)$ and $\partial_\eta = \frac{1}{2}(\partial_x + c^{-1}\partial_t)$, so

$$\frac{\partial^2 y}{\partial \xi \partial \eta} = -\frac{1}{4c^2} \left(\frac{\partial^2 y}{\partial t^2} - c^2 \frac{\partial^2 y}{\partial x^2} \right), \quad (189)$$

and the wave equation is exactly $\partial^2 y / \partial \xi \partial \eta = 0$. Integrate twice: $\partial y / \partial \eta$ is independent of ξ , hence a function of η alone; integrating it in η and adding the “constant” (a function of ξ),

$$\boxed{y(x, t) = f(x - ct) + g(x + ct)} \quad (190)$$

with f, g arbitrary twice-differentiable profiles — and this is *every* solution, by the construction just run. A right-moving and a left-moving shape, each translating rigidly at speed c (verify forward: each term is annihilated by one factor of the operator, by the chain rule).

The two faces are two bases for one solution space. Each standing wave is a pair of counter-propagating traveling waves,

$$\sin(kx) \cos(\omega t) = \frac{1}{2} \left[\sin(k(x - ct)) + \sin(k(x + ct)) \right] \quad (\omega = ck), \quad (191)$$

by the product-to-sum identity of §4.1. The mode basis fits fixed ends; the traveling basis fits pulses and the open line. (On the finite string the traveling picture survives as the method of reflections; not developed here [stated].)

9.6 The dictionary, and the arc completed

	Chain (N masses)	String (continuum)
configuration	vector $\vec{y} = (y_1, \dots, y_N)$	field $y(x, t)$
dynamics	$\ddot{\vec{y}} = -\mathbf{A}_N \vec{y}$	$\partial_t^2 y = c^2 \partial_x^2 y$
coupling	tridiagonal matrix $\mathbf{A}_N$	operator $-c^2 \partial_x^2$
second difference	$y_{j-1} - 2y_j + y_{j+1}$	$a^2 \partial_x^2 y$
boundary	phantom sites $y_0 = y_{N+1} = 0$	$y(0, t) = y(L, t) = 0$
mode label	$\theta_n = n\pi/(N+1)$, $n \leq N$	$k_n = n\pi/L$, n unbounded
mode shape	$v_j^{(n)} = \sin(j\theta_n)$	$\sin(k_n x)$
initial data	expansion in eigenvectors of $\mathbf{A}_N$	Fourier sine series
inner product	$\sum_j u_j v_j$	$\int_0^L u(x) v(x) dx$
dispersion	$\omega = 2\omega_0 \sin(\theta/2)$, ceiling $2\omega_0$	$\omega = ck$, no ceiling

System	What it added
SHO	the complex ansatz $e^{\lambda t}$; amplitude–phase in one number
damped	complex roots classify three regimes; Q from decay
forced (undamped)	particular + homogeneous; beats; resonance catastrophe
forced (damped)	$\hat{z} = f_0/(\omega_0^2 - \omega^2 + 2i\beta\omega)$; transients vs. steady state
resonance	peak at $\sqrt{\omega_0^2 - 2\beta^2}$; width 2β ; Q from linewidth
coupled	eigenvalue problem; normal coordinates decouple
forced coupled	one forced HO per mode; two resonances; antiresonance
N -mass chain	exponential ansatz <i>in space</i> ; dispersion relation
continuum limit	difference $\rightarrow$ derivative; the wave equation, $c^2 = T/\mu$
general solution	mode superposition; initial data by Fourier expansion

Every step reused the previous ones: a single ansatz — exponential in time, then again in space — one lemma about real parts, and one change of basis carry the subject from a mass on a spring to the wave equation.

10 Outlook: Waves in Two and Three Dimensions

Everything so far lived on a line. This closing section is an *introduction only*: we exhibit the higher-dimensional wave equation, indicate where it comes from, and point at what is new — the systematic treatment belongs to a later course.

Where the equation comes from. Repeat the bead construction on a plane: a square grid of masses with spacing a , tension T , each mass coupled to its *four* nearest neighbors, displacing by $u_{ij}(t)$ perpendicular to the grid. (We write u for the displacement field now that x and y are both needed as coordinates.) The transverse force analysis of §9.1 applies along each grid direction separately, so the two directions simply add:

$$m \ddot{u}_{ij} = \underbrace{\frac{T}{a} (u_{i+1,j} - 2u_{ij} + u_{i-1,j})}_{\text{second difference in } x} + \underbrace{\frac{T}{a} (u_{i,j+1} - 2u_{ij} + u_{i,j-1})}_{\text{second difference in } y}. \quad (192)$$

Each second difference becomes a^2 times a second partial derivative in the continuum limit, exactly as in (173), so

$$m \frac{\partial^2 u}{\partial t^2} = \frac{T}{a} \cdot a^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + O(a^4). \quad (193)$$

The quantities to hold fixed are the two-dimensional analogs of μ and T : the *areal density* $\sigma \equiv m/a^2$ (mass per area) and the *membrane tension* $\tau \equiv T/a$ (force per unit length of edge — in a sheet, tension pulls across lines, not at points). Dividing by $m = \sigma a^2$:

$$\frac{\partial^2 u}{\partial t^2} = \frac{Ta}{m} \nabla^2 u + O(a^2) = \frac{\tau}{\sigma} \nabla^2 u + O(a^2) \quad \implies \quad \boxed{\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u, \quad c^2 = \frac{\tau}{\sigma}} \quad (194)$$

where $\nabla^2 \equiv \partial_x^2 + \partial_y^2$, the *Laplacian*, is the continuum’s “sum of second differences over all directions.” The formula $c^2 = \tau/\sigma$ is the exact structural echo of $c^2 = T/\mu$: restoring agent per unit of geometry, over inertia per unit of geometry. In three dimensions the same equation with $\nabla^2 = \partial_x^2 + \partial_y^2 + \partial_z^2$ governs sound in air and, in a deeper development, light.

What survives, and what is new. Every structural element of this document transfers. The equation is linear: superposition holds. Fixed-boundary modes exist, oscillate independently, and are orthogonal; initial data is expanded in them by projection — Fourier’s idea, now in two variables. For a rectangular membrane ($0 \leq x \leq L_x$, $0 \leq y \leq L_y$, edges clamped), the mode shapes are products of the familiar one-dimensional ones, and one verifies by direct substitution that

$$u_{nm}(x, y, t) = \sin\left(\frac{n\pi x}{L_x}\right) \sin\left(\frac{m\pi y}{L_y}\right) \cos(\omega_{nm}t), \quad \omega_{nm} = c\pi \sqrt{\frac{n^2}{L_x^2} + \frac{m^2}{L_y^2}} \quad (195)$$

solves the equation with the correct boundary values (each factor contributes its $-k^2$; the Laplacian adds them; the frequency is c times the length of the wavevector (k_x, k_y)).

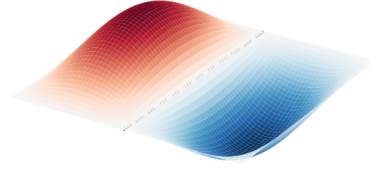
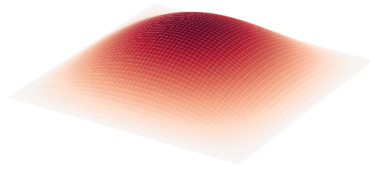
Three genuinely new phenomena are already visible in these formulas (and in Fig. 10):

- **Inharmonic overtones.** The frequencies ω_{nm} are *not* integer multiples of the lowest one — no analog of the harmonic series survives. This is audible: a string sings a pitch, a drum thuds.
- **Nodal lines.** A one-dimensional mode has nodal *points*; a membrane mode is silent along nodal *curves* (here the grid lines where either sine vanishes), the patterns made visible by sand in Chladni’s classic experiment.
- **Degeneracy.** On a square membrane ($L_x = L_y$), the modes (n, m) and (m, n) share a frequency: distinct motion patterns, identical pitch. This is no accident: reflecting the square across its diagonal ($x \leftrightarrow y$) maps solutions to solutions and interchanges the pair $(n, m) \leftrightarrow (m, n)$, so the two must share a frequency — degeneracy forced by the symmetry of the square, a first glimpse of the link between symmetry and degeneracy that becomes a central theme in quantum mechanics.

None of this requires new methods — an ansatz, a boundary condition, a projection — only richer geometry for them to act on. That is where the subject goes next.

$$(n, m) = (1, 1): \omega = 1.000 \omega_{11}$$

$$(n, m) = (2, 1): \omega = 1.581 \omega_{11}$$



$$(n, m) = (1, 2): \omega = 1.581 \omega_{11}$$

$$(n, m) = (2, 2): \omega = 2.000 \omega_{11}$$

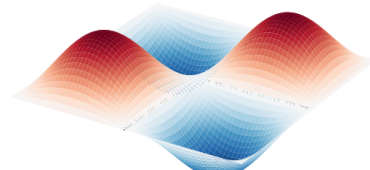
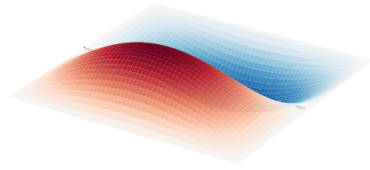


Figure 10: The four lowest modes of a square membrane ($L_x = L_y$), rendered as surfaces; dashed lines mark the interior nodal lines, where the membrane never moves. The frequency ratios $1 : \sqrt{5/2} : \sqrt{5/2} : 2$ display all three phenomena discussed below: they are not integer multiples of the fundamental, the nodes form curves rather than points, and the pair $(2, 1)$, $(1, 2)$ is degenerate.